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**Testing Market Efficiency: Evidence
from the US Dollar Index and
Emerging Markets**

BACHELOR THESIS

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Declaration on the Use of AI Tools

I confirm that artificial intelligence tools were used in the preparation of the submitted work. Specifically, they were used in the following ways:

- improvement of grammar and language style,
- searching for and summarizing literature,
- editing and expanding textual sections,
- generation of parts of source code,

Each use of any of the above methods is separately documented in an appendix to the thesis (see Appendix E), which provides a more detailed explanation of which tool was used in which specific part of the work; the author's own contribution is highlighted.

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- my work is original and was prepared in accordance with ethical standards, in particular the Code of Ethics of the Prague University of Economics and Business; that all information sources used are properly cited in the work; and that any generated content has been verified by me;
- I accept full responsibility for the entire content of the thesis.

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Abstrakt

Tato bakalářská práce se zabývá otázkou, zda je možné z finančních trhů systematicky získávat prediktivní informace, nebo zda se chovají efektivně ve smyslu slabé a polosilné formy hypotézy efektivních trhů (EMH). Pozornost je zaměřena na dva likvidní instrumenty z odlišných částí světového trhu: dolarový index (DXY), který reprezentuje vyspělý tržní prostor, a burzovně obchodovaný fond iShares MSCI Emerging Markets ETF (EEM), jenž slouží jako ukazatel vývoje na rozvíjejících se akciových trzích. Práce vytváří predikční rámec vycházející z ekonometrické analýzy časových řad a metod strojového učení a sleduje, nakolik lze zlepšit odhady výnosů oproti jednoduchému referenčnímu modelu, a systematicky modeluje podmíněnou volatilitu. Porovnání výsledků pro DXY a EEM ukazuje, jak míra vyspělosti a institucionální prostředí trhu ovlivňují limity predikovatelnosti a praktický prostor pro aktivní obchodní strategie.

Klíčová slova

Hypotéza efektivních trhů, strojové učení, analýza časových řad, rozvíjející se trhy, dolarový index, prediktivní modelování, modelování volatility.

Abstract

This bachelor thesis investigates whether financial markets contain exploitable predictive structure or are efficient in the weak and semi-strong forms of the Efficient Market Hypothesis (EMH). The analysis focuses on two liquid instruments that represent different segments of the global financial system: the U.S. Dollar Index (DXY), as a proxy for a mature developed market, and the iShares MSCI Emerging Markets ETF (EEM), as a benchmark for emerging equity markets. The thesis develops a forecasting framework that combines traditional time-series econometrics with modern machine learning techniques to assess the predictability of returns relative to a simple benchmark model, and to systematically model the conditional volatility. By comparing results across DXY and EEM, the study examines how market maturity and institutional quality shape the limits of predictability, and whether advanced predictive algorithms translate into economically meaningful gains for active trading strategies.

Keywords

Efficient Market Hypothesis, Machine Learning, Time Series Forecasting, Emerging Markets, US Dollar Index, Predictive Modelling, Volatility Modelling.

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List of abbreviations

ACF Autocorrelation Function	HML High Minus Low
ADF Augmented Dickey–Fuller Test	JB Jarque–Bera Test
AIC Akaike Information Criterion	JPY Japanese Yen
AMH Adaptive Markets Hypothesis	KDE Kernel Density Estimation
AR Autoregressive Model	KPSS Kwiatkowski–Phillips–Schmidt–Shin Test
ARCH Autoregressive Conditional Heteroskedasticity	Lasso Least Absolute Shrinkage and Selection Operator
ARIMA Autoregressive Integrated Moving Average	LM Lagrange Multiplier
ARIMAX Autoregressive Integrated Moving Average with Exogenous Variables	LRT Likelihood Ratio Test
ARX Autoregressive Model with Exogenous Variables	ML Machine Learning
CAD Canadian Dollar	MLE Maximum Likelihood Estimation
CAPM Capital Asset Pricing Model	MSCI Morgan Stanley Capital International
CART Classification and Regression Trees	MSE Mean Squared Error
CV Cross-Validation	OLS Ordinary Least Squares
DA Directional Accuracy	OOS Out-of-Sample
DM Diebold–Mariano Test	PACF Partial Autocorrelation Function
DS Difference-Stationary	Q-Q Quantile-Quantile
DXY US Dollar Index	RBF Radial Basis Function
EEM iShares MSCI Emerging Markets ETF	RF Random Forest
EMH Efficient Market Hypothesis	RMSE Root Mean Squared Error
ETF Exchange-Traded Fund	ROC Rate of Change
EUR Euro	S&P Standard & Poor’s
FX Foreign Exchange	SMB Small Minus Big
GARCH Generalized Autoregressive Conditional Heteroskedasticity	SVR Support Vector Regression
GBP British Pound Sterling	TS Trend-Stationary
HLN Harvey–Leybourne–Newbold Correction	USD United States Dollar
	VIX Volatility Index
	VR Variance Ratio Test
	WTI West Texas Intermediate
	XGBoost Extreme Gradient Boosting

Introduction

The publication of Fama's work in 1970 significantly reoriented the world of finance. The impact of this contribution was later recognised by the Nobel Committee, which awarded Fama the Sveriges Riksbank Prize in Economic Sciences in 2013. Fama put forward a provocative claim for his time: that markets cannot be systematically beaten over the long term, and any attempt to do so may only reduce investor returns in the future – a concept named the Efficient Market Hypothesis (EMH). This implies that active management on Wall Street provides no added value compared to passive tracking. This concept became the basis for the theoretical dominance of passive investment strategies.

In his work, Fama introduced three degrees of efficiency: weak, semi-strong, and strong. This paper focuses on the first two degrees, as the strong form is virtually impossible to test in standard frameworks, since it assumes that market participants have inside information.

The weak form tests whether an asset's historical prices can be used to predict its future behaviour. The semi-strong form expands the information space from the specific asset to all possible information publicly available to the investor. Both forms have been tested in practice for many years.

However, global financial markets are comprised of a vast number of diverse assets with significantly different properties, meaning that testing one specific asset cannot represent the entire market. To make the conclusions more robust, this thesis examines two assets that are economically connected but structurally distinct. These are the US Dollar Index (DXY) and the MSCI Emerging Markets ETF (EEM). The DXY represents a highly liquid, mature developed market. The EEM serves as a benchmark for developing equity markets. The comparison will show whether efficiency depends on market development and its properties.

Along with testing the weak and semi-strong forms of the EMH, this study also examines the predictability of volatility. Understanding and modelling the underlying risk is an integral part of financial analysis. It is an important tool for modern portfolio management that remains relevant and theoretically consistent with the efficient market hypothesis.

The remainder of the thesis is organised as follows. Chapter 1 outlines the theoretical foundations of market efficiency and reviews the relevant literature. Chapter 2 details the methodology, explaining the chosen econometric models, machine learning algorithms, and evaluation techniques. Chapter 3 introduces the dataset, feature engineering, and exploratory data analysis. Chapter 4 presents the empirical findings for both return predictability and volatility modelling, culminating in a comparative discussion. Finally, Chapter 5 serves as the conclusion, summarising the main takeaways and providing a final verdict on the efficiency of the studied markets.

1. Literature Review

1.1 Theoretical Foundations: Market Efficiency

The Efficient Market Hypothesis (EMH), formalised by Fama (1970), provides the central theoretical framework for studying asset price dynamics. Its core premise is that financial markets are informationally efficient – that is, asset prices fully reflect all available information. Fama established a taxonomy of efficiency based on the information set available to market participants, distinguishing between weak-form, semi-strong-form, and strong-form efficiency. The weak-form EMH holds that current security prices fully incorporate all historical price and return data, meaning that past price movements offer no systematic predictive value for future returns. The semi-strong form extends this condition to all publicly available information – the broader hypothesis whose empirical evaluation constitutes the central methodological challenge of this study. The strong form implies that prices reflect all existing information, including private or insider knowledge.

Testing the EMH is, however, complicated by the “joint-hypothesis problem” (Fama, 1991): any statistical test of market efficiency must be conducted jointly with an equilibrium asset-pricing model, making any rejection ambiguous by construction. The theoretical implications of this constraint, along with the selection of appropriate benchmarks for this study, are discussed in Section 1.2.

Under the weak-form EMH, early theoretical models generally assumed that stock prices follow a random walk strictly, implying that successive price changes are independent and identically distributed. Lo and MacKinlay (1988) were among the first to challenge this view directly, using a variance ratio test to reject the random walk hypothesis for weekly US stock market returns. Comparing variance estimators derived from data sampled at different frequencies, they found significant positive serial correlation in short-horizon returns – evidence that financial time series contain predictable components inconsistent with a pure random walk.

To reconcile the classical EMH with growing empirical anomalies and behavioural finance critiques, Lo (2004) proposed the Adaptive Markets Hypothesis (AMH). Drawing on Simon (1955)’s notion of bounded rationality and “satisficing”, the AMH applies principles from evolutionary biology – competition, adaptation, and natural selection – to financial markets. Within this framework, market efficiency is not a binary condition or a fixed equilibrium, but a context-dependent property that shifts over time and across markets. Profit opportunities emerge when environmental conditions change or when market participants make systematic errors, yet these opportunities erode as participants learn and adapt. Testing this across structurally different markets is therefore not incidental – it is a prerequisite for a meaningful empirical evaluation of the hypothesis.

1.2 The Joint-Hypothesis Problem and Asset-Pricing Benchmarks

Testing market efficiency requires a benchmark model of expected returns. Historically, the Capital Asset Pricing Model (CAPM) served as the primary such benchmark. However, to address systematic empirical anomalies that the CAPM failed to explain, Fama and French developed their seminal multi-factor models (Fama & French, 1992). The classic Fama-French framework expands the asset-pricing model by incorporating fundamental firm-specific characteristics, introducing factors such as SMB (Small Minus Big) and HML (High Minus Low) to capture the risk premia associated with a company's market capitalisation and its book-to-market equity ratio, respectively.

While the Fama-French models remain the most widely adopted benchmark for cross-sectional asset pricing and equity portfolio evaluation, they are incompatible with the predictive framework of this thesis for two reasons. First, the methodological objectives diverge. The Fama-French framework is an equilibrium pricing model designed to explain cross-sectional variation in expected returns among different portfolios over long horizons. This thesis, by contrast, focuses on out-of-sample time-series forecasting – predicting short-term asset directions and return magnitudes using exogenous macroeconomic variables and non-linear Machine Learning algorithms, not on explaining cross-sectional variation in expected returns.

Second, the Fama-French factors are simply incompatible with the assets studied here. They are derived from firm-level accounting fundamentals, yet the DXY is a macroeconomic currency basket. Fiat currencies carry no book value, corporate earnings, or market capitalisation, making equity-based fundamental factors inapplicable by construction. Furthermore, while the iShares MSCI Emerging Markets ETF (EEM) does consist of corporate equities, this study treats it strictly as a macroeconomic proxy for developing economies: a single time series driven by global shocks such as US Treasury yields, the VIX, and commodity cycles, rather than a portfolio to be decomposed along firm-level accounting dimensions.

1.3 Developed and Emerging Market Efficiency

In the empirical literature, market efficiency is not treated as a uniform property but as a relative one: the degree of predictability varies across markets depending on their institutional, regulatory, and structural characteristics (Lim & Brooks, 2011). This distinction is especially relevant when comparing highly liquid, mature markets with developing ones.

Developed financial markets, such as those underlying the US Dollar Index (DXY), are generally characterised by high liquidity, prices that adjust to new information quickly, and minimal frictions. These conditions push asset prices closer to a random walk, making structural inefficiencies hard to exploit. Emerging markets, such as those tracked by the iShares MSCI Emerging Markets ETF (EEM), operate under different conditions. Lim and Brooks (2011) identify factors such as limited corporate transparency, weak regulatory frameworks, and in-

adequate investor protection as persistent obstacles to achieving even semi-strong efficiency. As a result, empirical studies tend to find more pronounced anomalies and predictability in emerging economies than in their developed counterparts.

Emerging markets are also more sensitive to new information and prone to price deviations from fundamental values. Cajueiro and Tabak (2004) show that efficiency in emerging equity markets evolves over time and is positively related to market capitalisation and negatively related to average trading costs. Gazel (2020) similarly finds that while random walk models are the standard tool for weak-form testing, emerging markets often present a more complex picture, where structural barriers to information access prevent prices from moving entirely randomly.

The distinction is further reinforced by evidence from foreign exchange research. Meese and Rogoff (1983) showed that no structural macroeconomic model could consistently outperform a random walk in forecasting major currency exchange rates out of sample, suggesting that developed FX markets broadly conform to weak-form efficiency. Emerging market currencies, by contrast, have been shown to exhibit more persistent deviations from random walk behaviour, partly due to the same structural frictions that affect their equity markets (Cajueiro & Tabak, 2004). This asymmetry is precisely what this thesis aims to quantify.

This evidence suggests that market efficiency is a spectrum rather than a binary state. Bock and Geissel (2024) confirm that the magnitude and duration of market inefficiencies vary across regions and shift over time. Jointly examining a developed-market index and an emerging-market proxy therefore offers a more complete picture of where the limits of financial predictability actually lie.

1.4 Macroeconomic Linkages: DXY and Emerging Markets

Testing the semi-strong form of the EMH means asking whether publicly available macroeconomic and financial information can predict future asset returns (Fama, 1991). In practice, this means adding exogenous cross-asset variables to the forecasting models. Chen et al. (1986) showed that macroeconomic forces – such as interest rate structures, commodity prices, and broad market indices – act as systematic risk factors that influence asset pricing. This study follows that logic, moving beyond historical prices to include macroeconomic predictors on both the DXY and EEM sides.

The DXY, as a representative of developed foreign exchange markets, reflects the relative strength of the US currency and is closely tied to global macroeconomic conditions (Isard & Faruquee, 1998). The predictive models in this thesis incorporate key cross-asset instruments with established links to capital flows: US Treasury yields, major commodities such as oil, gold, and copper, and the S&P 500 index. The VIX is also included as a proxy for global market uncertainty and investor sentiment (Federal Reserve Bank of St. Louis, 2026), given how closely it tracks investor risk appetite.

These linkages are especially pronounced in emerging economies. The EEM, which captures equity performance across 24 emerging countries (MSCI, 2026), is used here as a proxy for that environment; a detailed description of both assets is provided in Chapter 3. Lim and Brooks (2011) note that emerging markets tend to exhibit higher volatility due to structural dependence on commodity exports, political instability, and sensitivity to exchange rate movements. Shifts in US interest rates and dollar strength have a direct effect on capital flows into these markets, making them especially exposed to global macroeconomic shocks.

Both the classical models (e.g., ARIMAX) and the Machine Learning algorithms include these variables. The central question is whether global macroeconomic information is absorbed into prices immediately, or whether it leaves predictable traces – signals that, if detectable, would constitute a violation of the semi-strong EMH.

1.5 Traditional and Machine Learning Approaches to Forecasting

The choice of forecasting methodology shapes how the EMH can be tested in practice. Among traditional approaches, models such as ARIMA, formalised by Box and Jenkins (1970), are valued for their parametric structure, interpretability, and well-established inferential framework. To handle the volatility clustering that is common in financial returns, the GARCH framework introduced by Bollerslev (1986) has become the standard tool for modelling conditional variance. These models are transparent and theoretically grounded, but their reliance on linearity assumptions limits their ability to capture more complex patterns in high-dimensional data (Masini et al., 2023).

Machine Learning algorithms – Random Forests, Support Vector Regressors, and Gradient Boosting – take a different route. They detect non-linear relationships and complex feature interactions without imposing parametric assumptions on the data (Sezer et al., 2020). This makes them well-suited for settings where the predictor set is large and the functional form of the relationships is unknown – precisely the case when testing the semi-strong EMH with a broad set of macroeconomic and cross-asset variables.

In this thesis, the two approaches are not treated as competing alternatives. Traditional models provide interpretable baselines and a rigorous statistical framework for inference, while ML algorithms explore non-linear structure that simpler models may overlook. Using both in parallel allows for a richer assessment of predictability – one that is robust to the choice of modelling approach.

This combination is necessary in part because ML flexibility carries a well-documented risk: models with high capacity can fit noise rather than genuine signal, particularly in financial time series where the signal-to-noise ratio is low (Makridakis et al., 2018). To address this, all ML forecasts are benchmarked against the traditional baselines using the Diebold-Mariano test (Diebold & Mariano, 1995), which formally assesses whether differences in predictive accuracy are statistically meaningful. A Permutation Test (Wasserman, 2004) is applied as

an additional safeguard: by randomly shuffling the temporal order of the target variable, it destroys any real conditional dependence between predictors and returns while leaving the marginal return distribution intact. If a model cannot outperform its permuted counterpart, its apparent accuracy is treated as spurious – a product of overfitting rather than genuine market inefficiency.

2. Methodology

2.1 Fundamentals of Time Series Analysis

A financial time series is a realisation of a stochastic process, defined as a sequence of random variables observed at discrete, equally spaced time intervals (Tsay, 2010). Time series analysis seeks to model the dynamic dependence of this sequence, often decomposing the data into unobserved components such as deterministic trends, periodic seasonality, and stochastic noise (Enders, 2015).

To mathematically represent the dynamic relationships between current and past observations, the lag operator L (often denoted as the backshift operator B) is widely employed. The lag operator shifts the time index of a variable backward by one period, defined as $Ly_t = y_{t-1}$ (Hamilton, 1994). By extension, applying the operator k times yields $L^k y_t = y_{t-k}$. This notation is useful for expressing complex autoregressive and moving average polynomials concisely.

In financial econometrics, raw asset prices P_t are often transformed prior to modelling to address their non-stationary nature and exponential growth. Prices are typically converted into continuously compounded (logarithmic) returns. Formally, the log return r_t is expressed as

$$r_t = \ln(P_t) - \ln(P_{t-1}) = (1 - L) \ln(P_t). \quad (2.1)$$

This transformation is adopted for several reasons (Tsay, 2010). First, unlike simple returns, which have a strict lower bound of -100% , log returns can take any value on the real line. This absence of a lower bound makes them more suitable for modelling with standard symmetric probability distributions. Second, multi-period returns can be easily calculated as the simple sum of single-period log returns. Furthermore, the logarithmic difference transformation is a standard method for eliminating unit-root non-stationarity in price levels, yielding a sequence that fluctuates around a constant mean.

2.2 Stationarity and Non-Stationarity

Time series forecasting and inference rely on the concept of stationarity, which requires that the statistical properties of the data-generating process remain constant over time. A process is strictly stationary if the joint probability distribution of any set of observations is invariant under a time shift (Tsay, 2010). Because strict stationarity is empirically difficult to verify, econometric analysis typically relies on weak stationarity (or covariance stationarity). A time series y_t is weakly stationary if it satisfies three conditions: a finite and constant mean $E(y_t) = \mu$, a constant variance $Var(y_t) = \sigma^2$, and an autocovariance $Cov(y_t, y_{t-k}) = \gamma_k$ that depends only on the lag length k and not on the specific time t (Hamilton, 1994; Tsay, 2010).

The concept of stationarity can be illustrated through the AR(1) process, which takes the form:

$$y_t = \phi_0 + \phi_1 y_{t-1} + \epsilon_t, \quad (2.2)$$

where ϵ_t is a white noise error term with zero mean and variance σ_ϵ^2 . The stability of this process is determined by the autoregressive coefficient ϕ_1 . The necessary and sufficient condition for the AR(1) process to be weakly stationary is that $|\phi_1| < 1$ (Enders, 2015). Under this condition, the effect of any past shock ϵ_{t-k} on the current value decays geometrically toward zero, and the series exhibits mean reversion.

If $\phi_1 = 1$, the process becomes a random walk, $y_t = y_{t-1} + \epsilon_t$, which is a standard example of a non-stationary unit root process (Enders, 2015). In this scenario, the variance of the series is time-dependent and approaches infinity ($t\sigma_\epsilon^2$), meaning there is no constant long-run mean to which the series returns. Furthermore, the impact of past shocks does not decay; instead, every shock has a permanent effect on the level of the series (Hamilton, 1994).

Within non-stationary processes, the literature distinguishes between trend-stationary (TS) and difference-stationary (DS) models. A TS process fluctuates around a deterministic time trend (e.g., $y_t = \alpha + \beta t + \epsilon_t$) and can be made stationary by de-trending (Enders, 2015). In contrast, a random walk is a DS process; its trend is stochastic rather than deterministic. For a DS series, de-trending is inappropriate and can result in spurious regressions; it must be transformed via differencing, $\Delta y_t = (1-L)y_t$, to achieve stationarity (Enders, 2015; Hamilton, 1994). Determining whether a financial series is TS or DS indicates whether market shocks represent temporary deviations from a long-term path or permanent structural shifts.

2.3 Unit Root Tests and Series Diagnostics

Prior to evaluating the Efficient Market Hypothesis, this study examines the statistical properties of the financial time series. To verify the stationarity conditions, unit root tests are employed. The Augmented Dickey–Fuller (ADF) test is based on estimating the auxiliary regression:

$$\Delta y_t = \alpha_0 + \gamma y_{t-1} + \sum_{i=1}^p \beta_i \Delta y_{t-i} + \epsilon_t, \quad (2.3)$$

where Δy_t is the first difference of the series, α_0 is a constant, p is the lag order, and ϵ_t is a white noise error term (Enders, 2015). The null hypothesis assumes that the data-generating process contains a unit root ($\gamma = 0$), which is tested against the alternative of stationarity ($\gamma < 0$). The ADF test is complemented by the Kwiatkowski–Phillips–Schmidt–Shin (KPSS) test, which reverses the testing framework by defining stationarity as the null hypothesis.

In addition to stationarity, diagnostic tests are applied to the return series and subsequent model residuals. To test for the presence of serial correlation, the Ljung–Box Q -statistic (Tsay, 2010) is defined as

$$Q(m) = T(T+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{T-k}, \quad (2.4)$$

where T is the sample size, m is the number of lags, and $\hat{\rho}_k$ is the sample autocorrelation at lag k . The test evaluates the joint null hypothesis that the first m autocorrelations are simultaneously equal to zero ($\rho_1 = \dots = \rho_m = 0$).

Because financial time series frequently exhibit volatility clustering, the ARCH-LM (Lagrange Multiplier) test is applied to test for conditional heteroskedasticity. The procedure involves regressing the squared demeaned returns (or squared residuals) $\hat{\epsilon}_t^2$ on their own q lagged values (Enders, 2015), yielding the regression equation:

$$\hat{\epsilon}_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \hat{\epsilon}_{t-i}^2 + v_t, \quad (2.5)$$

where v_t is a white noise error term. The null hypothesis posits that there are no autoregressive conditional heteroskedasticity effects, meaning that $\alpha_1 = \alpha_2 = \dots = \alpha_q = 0$.

Finally, the assumption of normality is assessed using the Jarque–Bera (JB) test, which compares the sample skewness ($\hat{S}$) and kurtosis ($\hat{K}$) of the series against the theoretical moments of a Gaussian distribution (Tsay, 2010). The test statistic is computed as:

$$JB = \frac{T}{6} \left(\hat{S}^2 + \frac{(\hat{K} - 3)^2}{4} \right). \quad (2.6)$$

Under the null hypothesis of a normal distribution, the JB statistic asymptotically follows a χ^2 distribution with two degrees of freedom. The results of these tests inform the model selection in the subsequent empirical analysis.

2.4 Weak-Form Efficiency Tests: Variance Ratio and Runs Test

Testing the weak-form Efficient Market Hypothesis involves evaluating whether historical price movements contain predictable patterns. To assess this, the study employs two primary statistical procedures: the parametric Variance Ratio test for linear serial dependence and the non-parametric Runs test for directional randomness.

The Variance Ratio (VR) test, introduced by Lo and MacKinlay (1988), is based on a property of the random walk process: the variance of its increments is linear with respect to the sampling interval. If a financial asset's logarithmic price $\ln(P_t)$ follows a random walk, the variance of a q -period return equals q times the variance of a single-period return. The variance ratio $VR(q)$ is defined as

$$VR(q) = \frac{Var(\ln(P_t) - \ln(P_{t-q}))}{q \cdot Var(\ln(P_t) - \ln(P_{t-1}))} = 1 + 2 \sum_{k=1}^{q-1} \left(1 - \frac{k}{q}\right) \hat{\rho}_k, \quad (2.7)$$

where $\hat{\rho}_k$ represents the k -th order sample autocorrelation coefficient of the first differences (returns). Under the null hypothesis of a random walk, $VR(q) = 1$. A ratio greater than one implies positive serial correlation (persistence or trending behaviour), while a ratio less than one indicates negative serial correlation (mean reversion). Because financial time series

frequently exhibit time-varying volatility, the standard test statistic must be adjusted. Lo and MacKinlay (1988) developed a heteroskedasticity-robust test statistic, denoted as $z^*(q)$, which asymptotically follows a standard normal distribution under the null hypothesis, permitting inference even in the presence of ARCH effects.

To complement the parametric VR test, the non-parametric Runs test (Wald–Wolfowitz test) is applied to evaluate the randomness of the sequence of price directions (Wald & Wolfowitz, 1940). This test is robust to non-normal return distributions and extreme outliers because it discards the magnitude of price changes and considers only their signs. Let a return sequence be classified into n_1 positive returns and n_2 negative returns, with the total number of non-zero observations being $N = n_1 + n_2$. A 'run' is defined as a consecutive sequence of identical signs, and let R denote the total number of observed runs. Under the null hypothesis that the sequence of positive and negative returns is independent, the expected number of runs $E(R)$ and its variance $Var(R)$ are given by:

$$E(R) = \frac{2n_1n_2}{N} + 1, \quad (2.8)$$

$$Var(R) = \frac{2n_1n_2(2n_1n_2 - N)}{N^2(N - 1)}. \quad (2.9)$$

For a sufficiently large sample size, the standardised test statistic Z_{runs} approximates a standard normal distribution:

$$Z_{runs} = \frac{R - E(R)}{\sqrt{Var(R)}}. \quad (2.10)$$

A statistically significant Z_{runs} indicates that the sequence of price changes exhibits either systematic clustering (fewer runs than expected) or over-alternation (more runs than expected). Either condition is inconsistent with the weak-form Efficient Market Hypothesis.

2.5 Data Preparation and Research Design

Multivariate econometric models and machine learning algorithms require temporally aligned feature matrices without missing values. At any given time t , the algorithms process a cross-sectional vector of features; if an observation is missing due to a market closure, the entire row must either be discarded or imputed to calculate cross-asset dependencies. To align the disparate trading calendars of the US Dollar Index (DXY) and the MSCI Emerging Markets ETF (EEM), the dataset is constructed in trading time (business days), omitting the temporal gaps corresponding to regular weekends.

Asynchronous national holidays are handled by imputing the missing observations using the forward-fill method. While this technique imposes zero returns for missing periods, which can theoretically dampen estimated volatility and distort cross-asset correlations (Tsay, 2010), its effect is likely negligible provided the fraction of imputed observations is low. The exact number of imputed observations and its empirical impact are detailed in Section 3.2. By ensuring that only historically available information is used, the forward-fill technique allows for

strict out-of-sample evaluation and prevents over-fitting, addressing the core methodological concerns raised by Makridakis et al. (2018).

To evaluate the forecasting models, the framework must account for time-series properties. Standard cross-validation techniques that rely on random shuffling are invalid for financial returns. Such randomisation disrupts the intrinsic temporal dependence and sequential structure of the observations (Tsay, 2010). Consequently, the dataset is chronologically partitioned into a training set (comprising the first 80% of observations) and a hold-out test set (the remaining 20%).

Furthermore, algorithms such as regularised regressions and Support Vector Machines require input features to be on a similar scale. Without scaling, variables with larger absolute magnitudes dominate the objective function and the distance metrics. Therefore, both the training and test sets are standardised. To prevent data leakage, the standardisation parameters are estimated exclusively on the training partition. Let $x_{i,t}$ denote the i -th feature at time t . The empirical mean $\hat{\mu}_i^{(train)}$ and standard deviation $\hat{\sigma}_i^{(train)}$ are computed over the training period T_{train} . The training data is scaled using these parameters. Subsequently, the exact same parameters are applied to scale the out-of-sample test data:

$$z_{i,t}^{(test)} = \frac{x_{i,t}^{(test)} - \hat{\mu}_i^{(train)}}{\hat{\sigma}_i^{(train)}}. \quad (2.11)$$

This standardisation procedure ensures that the test set remains unseen during the model fitting phase.

To construct the predictive framework, all exogenous macroeconomic and cross-asset variables are temporally aligned with the target variable. Formally, if the target variable to be predicted is the return at time t , denoted as r_t , the corresponding feature vector X_t contains information available at time t :

$$X_t = [x_{1,t}, x_{2,t}, \dots, x_{k,t}]^\top. \quad (2.12)$$

In addition to returns, 21-day and 63-day rolling standard deviations of returns are included as exogenous features to measure historical volatility. Consequently, the initial data sample is truncated to account for the 63-day warm-up period required to compute the longest rolling window, ensuring that all models are trained and evaluated on an identical time frame.

The feature matrix also includes discrete autoregressive lags of the target variable (e.g., lags 1, 2, 3, and 5). The dimensionality and composition of the final feature sets differ between the linear models and the machine learning algorithms; the exact specifications of these feature spaces are detailed in Section 3.7.

For model selection and hyperparameter tuning in the machine learning algorithms, this study employs an expanding-window cross-validation technique. Unlike standard k -fold cross-validation, this approach respects the temporal ordering of the data to prevent future-to-past information leakage. It creates consecutive training sets (e.g., $[1, t]$, $[1, t + k]$, $[1, t + 2k]$) and

validates them on the immediately following chronological blocks, preserving the temporal sequence of the observations during the grid search (Tsay, 2010).

The out-of-sample evaluation utilises a rolling one-step-ahead forecasting scheme. To account for changing market dynamics and structural parameter shifts – consistent with the Adaptive Markets Hypothesis (Lo, 2004) – all linear models (ARIMA, ARIMAX, Ridge, Lasso) and non-linear algorithms (Random Forest, XGBoost, SVR) are periodically refitted on an expanding training window every 21 trading days (approximately one month).

Given the computational cost of executing grid search cross-validation across approximately 50 evaluation windows, the hyperparameter tuning for the machine learning models is conducted exclusively on the initial in-sample training partition. The identified optimal hyperparameter configurations are then held constant. During the out-of-sample evaluation, the models are periodically refitted on an expanding historical window every 21 days using these fixed parameters. This design enables the models to dynamically adapt to recent data, ensuring a consistent benchmark comparison while maintaining computational feasibility.

2.6 The ARIMA Forecasting Framework and Baseline

This section describes the ARIMA and GARCH models used in the analysis. The ARIMA family is used both for return forecasting and as the EMH baseline, while the GARCH component models conditional volatility. An ARIMAX extension adds exogenous macroeconomic predictors to assess the semi-strong form of market efficiency.

The EMH Benchmark: ARIMA(0,0,0)

Under the weak-form EMH, asset returns reflect all historical price information, rendering future returns linearly unpredictable (Fama, 1970). Consequently, the baseline for this study is the ARIMA(0,0,0) specification, which models log-returns as a constant mean process with white noise deviations – implying that the underlying log-prices follow a random walk with drift (Enders, 2015). Formally, the return r_t is modelled as:

$$r_t = c + \epsilon_t, \quad (2.13)$$

where c is the constant unconditional mean and ϵ_t is a white noise error term. If the competing econometric or machine learning models fail to statistically outperform this baseline out-of-sample, the data remain consistent with market efficiency.

Univariate Autoregressive Modelling: ARIMA(p,d,q)

To test for the presence of linear autocorrelation structure in returns, a general Autoregressive Integrated Moving Average (ARIMA) model is estimated. Since the financial price series are transformed into stationary logarithmic returns prior to modelling, the integration order is zero ($d = 0$), reducing the model to an ARMA(p,q) specification. The conditional mean

equation is defined as:

$$r_t = c + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t. \quad (2.14)$$

The autoregressive (p) and moving average (q) lag orders are selected algorithmically by minimising the Akaike Information Criterion (AIC). If the selected order deviates from (0,0,0), it formally indicates the presence of an internal linear structure in the asset's history.

Volatility Modelling: GARCH(1,1)

While the ARIMA model filters the conditional mean, empirical financial time series exhibit volatility clustering and non-constant variance – a phenomenon confirmed by the ARCH-LM tests in Chapter 3. To accommodate this conditional heteroskedasticity, the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model introduced by Bollerslev (1986) is employed.

This study utilises the parsimonious GARCH(1,1) specification. In most financial applications, lower-order GARCH models are sufficient to capture volatility clustering dynamics without suffering from over-parameterisation (Tsay, 2010). Furthermore, while a standard GARCH model with Gaussian errors generates some excess kurtosis relative to the normal distribution, this is often insufficient to fully capture the leptokurtic nature of financial returns. Therefore, to account for the heavy tails characteristic of empirical market data, the innovations are modelled using a standardised Student's t -distribution, following the established methodology of Bollerslev (1987).

The Joint ARIMA-GARCH Framework

To model both the conditional mean and conditional variance, this study integrates the two aforementioned models into a joint ARIMA-GARCH framework. The specification of the combined model is given by the following system of equations:

$$r_t = c + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \epsilon_t, \quad (2.15)$$

$$\epsilon_t = \sigma_t z_t, \quad z_t \sim t(\nu), \quad (2.16)$$

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2, \quad (2.17)$$

where $\omega > 0$, $\alpha_1 \geq 0$, and $\beta_1 \geq 0$ to ensure a positive conditional variance σ_t^2 , and z_t follows a Student's t -distribution with ν degrees of freedom.

Following the two-pass estimation method (Tsay, 2010), the parameters are estimated sequentially. First, the ARIMA($p,0,q$) model (2.14) is estimated to extract the mean-zero residuals ϵ_t . Second, the GARCH(1,1) model (2.16)–(2.17) is fitted directly to these extracted residuals. While joint maximum likelihood estimation is theoretically optimal, Tsay (2010) acknowledges that the sequential approach lacks formal asymptotic justification, though it tends to perform well in large samples. Given that the empirical datasets in this study encompass over 5,250 observations per asset, this two-pass procedure is employed due to the computational constraints of the Python implementation.

Testing Semi-Strong Efficiency: ARIMAX-GARCH

To evaluate the semi-strong form of market efficiency, the univariate framework is expanded to incorporate exogenous macroeconomic information. The model is augmented to an ARIMAX specification. Let $\mathbf{X}_{t-1}$ be the vector of exogenous cross-asset predictors shifted backward by one day to prevent look-ahead bias, and β be the corresponding coefficient vector. The mean equation of the ARIMAX-GARCH specification is:

$$r_t = c + \sum_{i=1}^p \phi_i r_{t-i} + \sum_{j=1}^q \theta_j \epsilon_{t-j} + \beta^\top \mathbf{X}_{t-1} + \epsilon_t. \quad (2.18)$$

The residuals from this expanded mean equation are subsequently modelled using the GARCH volatility equation. By comparing the out-of-sample predictive accuracy of the ARIMAX-GARCH model against the ARIMA(0,0,0) baseline, this study tests whether macroeconomic variables provide out-of-sample predictive power, which would constitute a deviation from semi-strong market efficiency.

2.7 Regularised Regression: Ridge and Lasso

Penalized regressions, such as Ridge and Lasso, retain the linear structure and interpretability of Ordinary Least Squares (OLS) while incorporating a regularisation mechanism. This regularisation addresses the limitations of OLS in high-dimensional feature spaces with highly correlated covariates, making these methods applicable to macroeconomic and financial forecasting (Masini et al., 2023).

When evaluating the semi-strong form of market efficiency, forecasting models incorporate macroeconomic data. While the ARIMAX model processes a concise set of base regressors, unpenalised linear models estimated via OLS are prone to overfitting in noisy environments with many predictors (Hastie et al., 2009). Regularised (or shrinkage) regressions extend the ARIMAX framework by restricting the parameter space, which introduces a small bias but reduces the variance of the estimator. Furthermore, these regularisation techniques can identify the most relevant variables, revealing the underlying structure of the information processing. Although this feature selection does not serve as a direct proof of market efficiency, it warrants consideration.

Ridge regression is considered first, before the Lasso specification. It imposes an L_2 penalty on the squared magnitude of the model parameters. The Ridge estimator is defined as the solution to the following optimisation problem:

$$\hat{\beta}^{Ridge} = \arg \min_{c, \beta} \left\{ \sum_{t=1}^T (r_t - c - \beta^\top \mathbf{X}_{t-1})^2 + \alpha \sum_{j=1}^k \beta_j^2 \right\}, \quad (2.19)$$

where r_t is the asset return, c is the intercept, $\mathbf{X}_{t-1}$ is the vector of k exogenous predictors, and α is the regularisation hyperparameter. By shrinking the coefficients, Ridge restricts the parameter space and handles highly correlated predictors (Hastie et al., 2009).

While Ridge shrinks coefficients, it does not perform feature selection, meaning all variables are retained in the final model. If the true underlying model is sparse – meaning only a subset of macroeconomic factors explains asset returns – the Lasso (Least Absolute Shrinkage and Selection Operator) regression is applicable (Tibshirani, 1996). Lasso applies an L_1 norm penalty:

$$\hat{\beta}^{Lasso} = \arg \min_{c, \beta} \left\{ \sum_{t=1}^T \left(r_t - c - \beta^\top \mathbf{X}_{t-1} \right)^2 + \alpha \sum_{j=1}^k |\beta_j| \right\}. \quad (2.20)$$

Because of the nature of the L_1 constraint, Lasso forces the weights of some features exactly to zero, thereby performing continuous shrinkage and automatic variable selection simultaneously (Tibshirani, 1996).

Both Ridge and Lasso are implemented to compare their predictive accuracy and to evaluate the trade-off between Ridge’s shrinkage under multicollinearity and Lasso’s sparse feature selection. The predictive capability of both regularised models depends on the regularisation strength hyperparameter, α . To prevent look-ahead bias, the optimal α for each model is determined through a chronological expanding-window cross-validation, applied exclusively within the training data, as outlined in Chapter 2.5.

2.8 Non-Linear Machine Learning Models

While regularised linear models successfully address high-dimensionality and multicollinearity, they assume that the relationship between macroeconomic variables and future asset returns is linear. However, empirical financial time series are frequently characterised by structural breaks, asymmetric responses to shocks, and non-linear interactions between global market factors (Nelson, 1991; Tsay, 2010). To capture these dependencies without imposing rigid parametric forms, this study employs three distinct non-linear machine learning algorithms: Random Forest, Extreme Gradient Boosting (XGBoost), and Support Vector Regression (SVR).

2.8.1 Random Forest

Introduced by Breiman (2001), a Random Forest is an ensemble learning method that constructs a collection of unpruned decision trees during the training phase. Let $\mathbf{X}_{t-1}$ be the vector of lagged exogenous features, and let $\hat{r}_t$ denote the predicted return at time t . The final prediction of the forest is the average of the predictions of K individual trees:

$$\hat{r}_t = \frac{1}{K} \sum_{k=1}^K h_k(\mathbf{X}_{t-1}, \Theta_k), \quad (2.21)$$

where h_k is the k -th regression tree and Θ_k represents an independently sampled random vector dictating the bootstrap sample and the random subset of features evaluated at each node split (Breiman, 2001).

Application to Financial Time Series: Single decision trees are highly sensitive to the noise inherent in financial data, frequently leading to overfitting. Random Forests mitigate this vulnerability through bagging (bootstrap aggregating) combined with feature subsampling. By decorrelating the individual trees and averaging their outputs, the algorithm reduces model variance (Hastie et al., 2009). This mechanism makes the Random Forest robust to the low signal-to-noise ratio of daily market fluctuations, allowing it to capture non-linear macroeconomic dependencies that classical econometric models might fail to detect.

2.8.2 Extreme Gradient Boosting (XGBoost)

While Random Forest builds trees independently in parallel to reduce model variance, Gradient Boosting builds them sequentially, with each new tree fitted to the pseudo-residuals of the previous iterations (Friedman, 2001). Formalised by Chen and Guestrin (2016), XGBoost is a scalable tree boosting system that optimises a regularised objective function. Let $f_k \in \mathcal{F}$ be a regression tree function from the space of all possible CART (Classification and Regression Trees), mapping the input features to a continuous leaf weight. At iteration k , the algorithm adds the tree function f_k to minimise:

$$\mathcal{L}^{(k)} = \sum_{t=1}^T l\left(r_t, \hat{r}_t^{(k-1)} + f_k(\mathbf{X}_{t-1})\right) + \Omega(f_k), \quad (2.22)$$

where l is a differentiable convex loss function measuring the difference between the target return r_t and the current prediction $\hat{r}_t^{(k-1)}$. The term $\Omega(f_k)$ is the penalty controlling tree complexity:

$$\Omega(f_k) = \gamma M_k + \frac{1}{2} \lambda \|w\|^2 + \alpha \|w\|_1, \quad (2.23)$$

where M_k represents the number of terminal leaves in the k -th tree, w is the vector of leaf weights, γ specifies the minimum loss reduction required to make a further partition, and λ and α govern the L_2 and L_1 regularisation strengths, respectively (Chen & Guestrin, 2016).

Application to Financial Time Series: As discussed previously, financial markets often exhibit asymmetric responses to shocks (Nelson, 1991). By sequentially fitting new trees to the gradient of the loss function, XGBoost can adapt to these non-linear dynamics. Furthermore, the L_1 and L_2 regularisation embedded into the objective function restricts the algorithm's complexity and explicitly reduces the risk of overfitting (Chen & Guestrin, 2016).

2.8.3 Support Vector Regression (SVR)

Support Vector Regression (SVR), an extension of the support vector framework originally proposed by Cortes and Vapnik (1995), approaches continuous forecasting by constructing an ϵ -insensitive “tube” around the training data. Instead of penalising all prediction errors like classical Ordinary Least Squares (OLS), SVR ignores errors that fall within a specified distance ϵ from the true target return r_t . The objective is to fit instances within this tube

while restricting the complexity of the model. Formally, this is expressed as the following optimisation problem:

$$\arg \min_{\mathbf{w}, b, \xi, \xi^*} \left\{ \frac{1}{2} \|\mathbf{w}\|^2 + C \sum_{t=1}^T (\xi_t + \xi_t^*) \right\}, \quad (2.24)$$

subject to the constraints that the residuals $r_t - (\mathbf{w}^\top \mathbf{X}_{t-1} + b)$ are bounded by ϵ and the non-negative slack variables ξ_t, ξ_t^* (Hastie et al., 2009). Here, the term $\frac{1}{2} \|\mathbf{w}\|^2$ serves as a regularisation mechanism that restricts the parameter space, b is the unpenalised bias term, and C is a regularisation hyperparameter. The parameter C dictates the trade-off between the generalisation capability of the model and the tolerance for prediction errors larger than ϵ .

To capture non-linear dependencies, SVR employs the kernel trick. The input features are mapped into a high-dimensional feature space using a kernel function $K(\mathbf{X}_i, \mathbf{X}_j) = \phi(\mathbf{X}_i)^\top \phi(\mathbf{X}_j)$ (Cortes & Vapnik, 1995). In this study, the Gaussian Radial Basis Function (RBF) kernel is utilised:

$$K(\mathbf{X}_i, \mathbf{X}_j) = \exp \left(-\gamma \|\mathbf{X}_i - \mathbf{X}_j\|^2 \right), \quad (2.25)$$

where $\mathbf{X}_i$ and $\mathbf{X}_j$ are the feature vectors of two distinct training observations (e.g., at two different time steps), and γ is a hyperparameter defining the radius of influence of a single training example.

Application to Financial Time Series: SVR is applicable to financial forecasting for two main reasons. First, the ϵ -insensitive loss function reduces the sensitivity of the estimator to extreme market outliers (Makridakis et al., 2018), as errors within the ϵ -margin are inherently not penalised. Second, the RBF kernel allows the algorithm to adapt to non-linear cross-asset dynamics without explicitly specifying their functional form (Cortes & Vapnik, 1995).

2.9 Hyperparameter Optimisation: Grid Search

Machine learning algorithms are parametrised not only by the internal weights learned during training, but also by hyperparameters. Hyperparameters, or tuning parameters, are structural settings defined prior to the learning process that control the complexity of the model (Hastie et al., 2009). Examples include the number of estimators and maximum tree depth in Random Forest and XGBoost, or the penalty parameter C , kernel coefficient γ , and tube width ϵ in Support Vector Regression. Because the optimal hyperparameter configuration is dataset-dependent, a systematic optimisation procedure is essential to minimise out-of-sample predictive error and prevent both underfitting and overfitting.

To identify the optimal parameters, this study employs the Grid Search methodology. Grid Search performs a search over a manually specified, discrete subspace of the hyperparameter space. For a given machine learning algorithm, let $\Lambda = \Lambda_1 \times \Lambda_2 \times \cdots \times \Lambda_k$ denote the Cartesian product of k predefined hyperparameter sets. The Grid Search algorithm systematically trains and evaluates a model for every single combination $\lambda \in \Lambda$.

To prevent data leakage and look-ahead bias, the evaluation of each parameter combination λ is conducted using the expanding-window cross-validation technique (Time Series Split) detailed in Chapter 2.5. For each fold, the model is trained on the expanding historical window and evaluated on the subsequent chronological validation block. The performance of each combination λ is quantified using the Mean Squared Error (MSE).

The hyperparameter configuration that minimises the average validation error across all temporal folds is selected as the optimal set. Once the optimal hyperparameters are identified, the final model is refitted on the entire in-sample training partition using this configuration. This optimised model is then deployed to generate out-of-sample forecasts on the hold-out test set. This tuning procedure is applied independently to the Random Forest, XGBoost, and SVR algorithms, ensuring that the benchmark comparison against the Efficient Market Hypothesis is conducted using optimally configured non-linear models.

2.10 Model Evaluation and Hypothesis Validation

To thoroughly assess the out-of-sample forecasting performance of the employed models and translate the economic theory of the Efficient Market Hypothesis into a testable empirical framework, a statistical evaluation design is required. Relying solely on standard forecast error metrics, such as the Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE), is insufficient in financial time series, as minor variations in forecast accuracy may be purely driven by sample noise (Makridakis et al., 2018). Therefore, this thesis formally defines and tests three primary pairs of statistical hypotheses:

- **Hypothesis 1: Weak-Form Market Efficiency**

H_0 : The asset return series is serially uncorrelated, and the sequence of price changes is purely random.

H_1 : The asset return series exhibits significant temporal dependence.

- **Hypothesis 2: Semi-Strong-Form Market Efficiency**

H_0 : The inclusion of exogenous macroeconomic and cross-asset variables does not significantly improve out-of-sample predictive accuracy compared to the baseline.

H_1 : The inclusion of exogenous variables significantly improves out-of-sample predictive accuracy.

- **Hypothesis 3: Volatility Modelling (Homoskedasticity vs. ARCH Effects)**

H_0 : The conditional variance of the asset returns is constant over time (homoskedasticity), and the squared residuals exhibit no serial correlation. This implies that market volatility cannot be predicted using historical information.

H_1 : The conditional variance of the asset returns is time-varying and exhibits volatility clustering (conditional heteroskedasticity), suggesting that the magnitude of market volatility contains predictable structure that may be captured by a GARCH-type model.

To empirically test these hypotheses, the out-of-sample point estimates are evaluated using formal statistical tests. Specifically, the weak-form efficiency hypothesis is evaluated using

the Variance Ratio test (Lo & MacKinlay, 1988) to detect linear autocorrelations and the non-parametric Runs test to assess directional randomness (Wald & Wolfowitz, 1940). Additionally, the modified Diebold-Mariano test (Diebold & Mariano, 1995; Harvey et al., 1997) and the Binomial test are used to compare the accuracy and directional predictability of univariate ARIMA forecasts against the ARIMA(0,0,0) baseline. The semi-strong-form efficiency hypothesis is evaluated using a dual approach: the modified Diebold-Mariano and Binomial tests assess the predictive superiority of the ARIMAX and machine learning models over the baseline, while a block permutation test is employed to control for spurious data-mining patterns. Finally, the volatility modelling hypothesis is assessed using the ARCH-LM test on raw squared returns to detect the presence of conditional heteroskedasticity, and the Ljung-Box Q -statistic on squared standardised residuals to verify the adequacy of the fitted volatility models (Tsay, 2010).

Let N denote the total number of out-of-sample predictions. Let r_t represent the true logarithmic return at time t and $\hat{r}_t$ denote the forecasted return. The primary evaluation metric utilised for point forecasts is the RMSE, defined as:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum_{t=1}^N (\hat{r}_t - r_t)^2}. \quad (2.26)$$

While RMSE quantifies the average magnitude of the forecast error, it does not necessarily reflect the economic utility of a model. In financial forecasting, predicting the correct direction of the market is often of greater practical interest than minimising the absolute deviation (Tsay, 2010). Thus, Directional Accuracy (DA) is calculated to measure the proportion of times the model correctly anticipates the market's direction:

$$\text{DA} = \frac{1}{N} \sum_{t=1}^N \mathbf{1}_{\{\text{sgn}(\hat{r}_t) = \text{sgn}(r_t)\}}, \quad (2.27)$$

where $\mathbf{1}$ is the indicator function. To determine whether the directional accuracy statistically significantly differs from a random guess, a two-sided Binomial test is employed. Under the null hypothesis $H_0 : p = 0.5$ of random directional guessing, the number of correct predictions $X = \sum_{t=1}^N \mathbf{1}_{\{\text{sgn}(\hat{r}_t) = \text{sgn}(r_t)\}}$ follows a binomial distribution $B(N, 0.5)$. For a sufficiently large out-of-sample size N , the standardised test statistic is approximated by a standard normal distribution:

$$Z_{bin} = \frac{X - 0.5N}{\sqrt{N \cdot 0.5 \cdot 0.5}} \sim N(0, 1). \quad (2.28)$$

The null hypothesis is rejected in favour of the two-sided alternative $H_1 : p \neq 0.5$ if the resulting p -value is strictly below the standard significance level of $\alpha = 0.05$.

To evaluate the predictive accuracy of the competing models relative to the ARIMA(0,0,0) baseline, this study utilises the Diebold and Mariano (1995) (DM) test. Let $e_{1,t}$ and $e_{2,t}$ be the out-of-sample forecast errors of the baseline and the competing model, respectively. Using the squared error loss function, the loss differential is defined as $d_t = e_{1,t}^2 - e_{2,t}^2$. The null hypothesis of the DM test assumes equal predictive accuracy, meaning the expected loss

differential is zero ($H_0 : E[d_t] = 0$). The sample mean of the loss differential is $\bar{d} = \frac{1}{N} \sum_{t=1}^N d_t$, and its sample variance is estimated as $\hat{\sigma}_d^2$. The standard DM statistic is defined as:

$$DM = \frac{\bar{d}}{\sqrt{\hat{\sigma}_d^2/N}}. \quad (2.29)$$

Because financial forecast errors exhibit non-normality and the standard DM test can be oversized in finite samples, this study employs the modified Diebold-Mariano test proposed by Harvey et al. (1997) (HLN correction). For an h -step-ahead forecast, the HLN modification applies a sample-size correction factor to the standard DM statistic:

$$MDM = \sqrt{\frac{N+1-2h+h(h-1)/N}{N}} \cdot DM. \quad (2.30)$$

Since this research employs a rolling one-step-ahead forecasting design ($h = 1$), the correction factor simplifies to $\sqrt{(N-1)/N}$. To ensure robustness against heavy-tailed financial errors, the MDM statistic is compared against critical values from the Student's t -distribution with $N-1$ degrees of freedom, rather than the standard normal distribution (Harvey et al., 1997).

Finally, to control for overfitting and data-mining biases in Machine Learning algorithms, this study employs a non-parametric block permutation test. Under the null hypothesis of market efficiency, the target return y_t is independent of the lagged feature vector $\mathbf{X}_{t-1}$. While the standard independent and identically distributed (i.i.d.) permutation test (Wasserman, 2004) is widely used in statistics, it breaks the temporal dependence structure of financial returns. As noted by Diebold and Mariano (1995), standard randomisation tests are invalid for serially correlated data because arbitrary rearrangements destroy the underlying temporal dynamics.

To simulate the null distribution while addressing this limitation, the randomisation is conducted using block resampling. As Masini et al. (2023) emphasise, accounting for the time-dependence structure during resampling is important in time-series machine learning. The training target vector y_{train} is divided into non-overlapping blocks of 21 trading days. This block size was selected to capture monthly market cycles, consistent with the $\mathcal{O}(n^{1/3})$ asymptotic guideline for optimal block length. These blocks are randomly shuffled to generate a permuted surrogate vector, y_{train}^* . This transformation severs the conditional dependence between the exogenous variables and the target series while preserving the marginal distribution, short-run autocorrelation, and volatility clustering within each block.

The selected model is fitted to the randomised dataset $(\mathbf{X}_{\text{train}}, y_{\text{train}}^*)$ and evaluated on the unpermuted test set $\mathbf{X}_{\text{test}}$ to obtain the permuted out-of-sample error, denoted as $RMSE^*$. By repeating this procedure 200 times, an empirical null distribution of $RMSE^*$ is constructed. The null hypothesis of no predictability is rejected if the true out-of-sample RMSE of the model falls in the extreme left tail of the permuted $RMSE^*$ distribution (i.e., a p -value < 0.05). By preserving the time-series structure (Masini et al., 2023), this approach provides a statistically valid, distribution-free safeguard against spurious predictability.

3. Data and Exploratory Analysis

3.1 Object of Study and Data Sources

This study compares market efficiency across two distinct financial environments: developing economies and the global reserve currency. The developing markets are represented by the iShares MSCI Emerging Markets ETF (ticker: EEM), a diversified instrument capturing large and mid-cap equities across multiple emerging market countries. The developed market benchmark is the US Dollar Index (ticker: DX-Y.NYB), a basket tracking the value of the US Dollar relative to major global currencies. This pair was selected to contrast a structurally heterogeneous equity environment against a highly liquid currency index.

All historical daily price data were sourced from Yahoo Finance (2026), covering an approximately 21-year period from February 2005 to March 2026. To avoid information loss, the independently estimated univariate models retain their natural business-day calendars, comprising 5,326 trading days for DXY and 5,321 for EEM. For comparative visualisations and cross-asset correlations, the two series are intersected to a common timeline of exactly 5,319 joint trading days. This horizon spans multiple market regimes, including the 2008 Global Financial Crisis and the 2020 COVID-19 shock.

Figure 3.1: Cumulative Return Index: EEM and DXY on a dual y-axis (2005–2026)



Source: author's own calculations.

As shown in Figure 3.1, both the EEM and DXY indices exhibit overall cumulative growth over the sample period. While the EEM experiences substantial drawdowns, the DXY displays more cyclical trends, including a distinct appreciation phase starting in mid-2014. Despite this shared long-term growth, a pronounced inverse relationship exists between the two assets: systemic market crises consistently coincide with a flight to the US Dollar and simultaneous equity market declines in emerging markets.

3.2 Calculation of Returns

In financial econometrics, raw asset prices P_t are rarely modelled directly due to their non-stationary nature. A non-stationary time series, such as a random walk, possesses a stochastic trend; its variance is time-dependent and approaches infinity as time progresses (Enders, 2015). This violates the fundamental assumptions of classical forecasting models. To evaluate the stationarity of the data, the Augmented Dickey-Fuller (ADF) and Kwiatkowski-Phillips-Schmidt-Shin (KPSS) tests were applied to the raw price levels of both indices.

As summarised in Table 3.1, the tests confirm the presence of a stochastic trend. For both the US Dollar Index (DXY) and the MSCI Emerging Markets ETF (EEM), the ADF test yields high p -values, meaning the null hypothesis of a unit root cannot be rejected. Simultaneously, the KPSS test yields p -values below 0.01, rejecting the null hypothesis of stationarity.

Table 3.1: Stationarity Tests on Raw Price Levels (P_t)

Asset	ADF Stat	ADF p -value	KPSS Stat	KPSS p -value
EEM	-1.625	> 0.10	8.069	< 0.001
DXY	-1.497	> 0.10	8.813	< 0.001

Note. The ADF test was estimated with a constant and lag length selected by the Akaike Information Criterion (AIC), with a maximum lag of $L = \lceil 12(n/100)^{1/4} \rceil$. The KPSS test was applied with bandwidth selected automatically via the Newey–West procedure following Hobijn et al. (2004). A 5% significance level is used for both tests.

Source: author’s own calculations.

Subsequently, the tests were applied to the first differences. The iterative testing procedure determined the integration order to be $d = 1$ for both assets, as taking exactly one difference induced stationarity. These results indicate that the raw price series are consistent with non-stationary $I(1)$ processes. Direct predictive modelling on these levels would violate covariance stationarity and could result in spurious regressions.

To address this non-stationarity and achieve the covariance stationarity required for the forecasting frameworks, the daily closing prices are transformed into continuously compounded (logarithmic) returns. These are calculated as the first difference of the natural logarithm of prices: $r_t = \ln(P_t/P_{t-1})$. Logarithmic returns are preferred over simple percentage changes because they are time-additive, mathematically symmetric, and exhibit well-behaved statistical properties for econometric modelling (Tsay, 2010).

Table 3.2 summarises the test results for the transformed log-returns of EEM and DXY. The ADF test statistics (-15.970 and -22.044, respectively) reject the null hypothesis of a unit root at the 0.01 significance level. At the same time, the KPSS test statistics (0.0616 and 0.0416) are well below the critical thresholds. Due to the interpolation limits of the statistical software’s look-up tables, the exact p -values for the KPSS statistics cannot be computed. Instead, they can only be reported as exceeding the 0.10 level. Consequently, the null hypothesis of stationarity cannot be rejected, indicating that the logarithmic transformation

successfully eliminates the unit root.

Table 3.2: Stationarity Tests on Daily Log>Returns (r_t)

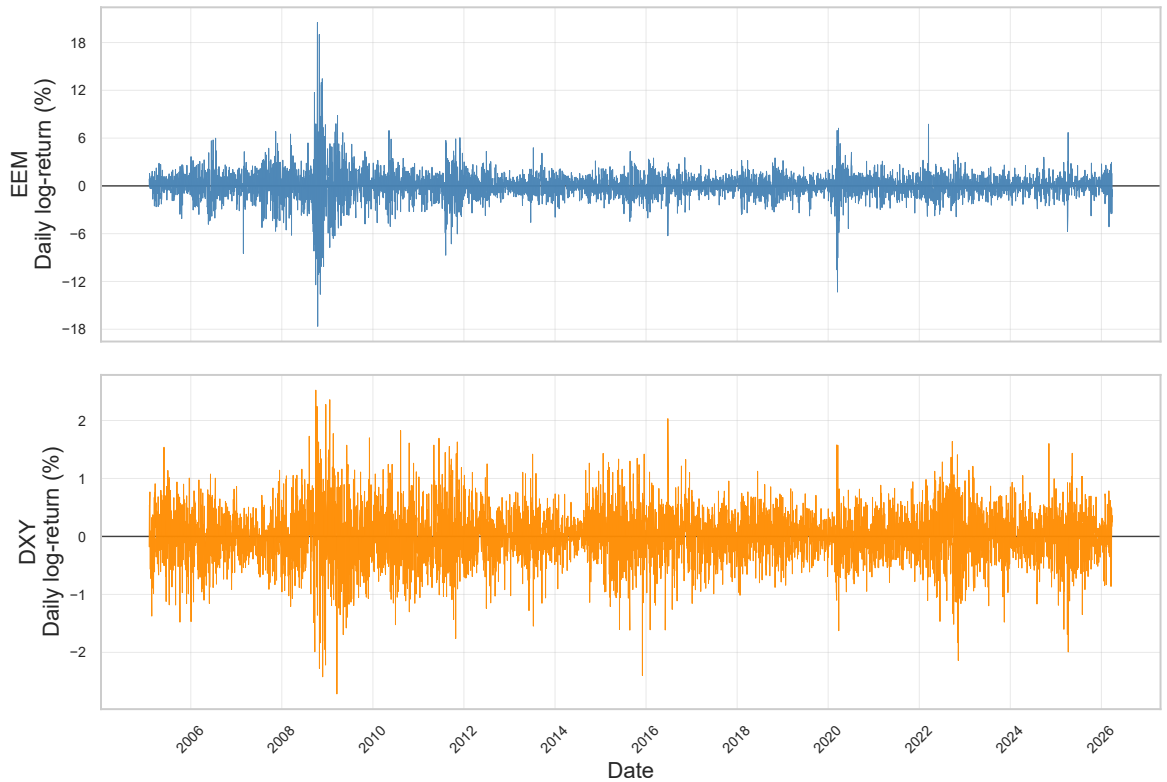
Asset	ADF Stat	ADF p -value	KPSS Stat	KPSS p -value
EEM	-15.970	< 0.001	0.0616	> 0.10
DXY	-22.044	< 0.001	0.0416	> 0.10

Note. The ADF test was estimated with a constant and lag length selected by the Akaike Information Criterion (AIC), with a maximum lag of $L = \lceil 12(n/100)^{1/4} \rceil$. The KPSS test was applied with bandwidth selected automatically via the Newey–West procedure following Hobijn et al. (2004). A 5% significance level is used for both tests.

Source: author’s own calculations.

The resulting stationary daily log-return series for both primary assets, presented in Figure 3.2, exhibit characteristic volatility clustering visible across the full sample period.

Figure 3.2: Daily Log>Returns: EEM vs DXY (2005–2026)



Source: author’s own calculations.

Both series exhibit volatility clustering, with periods of increased variance concentrated around the 2008 Global Financial Crisis and the 2020 COVID-19 shock. This log-return transformation is applied uniformly to all datasets in this study, encompassing both the target indices and the full set of exogenous variables.

Data alignment and forward-fill imputation procedures follow the research design outlined in Section 2.5. For the EEM dataset, aligning the exogenous cross-asset features to the 5,321-day

target calendar required forward-fill imputation on 44 trading days (approximately 0.83% of the sample). For the DXY dataset, aligning the exogenous features to the 5,326-day calendar required imputation on 49 trading days (approximately 0.92% of the sample). Given the low proportion of imputed observations for both datasets, any potential dampening effect on estimated volatility (Tsay, 2010) is unlikely to be of practical consequence.

3.3 Statistical Properties of Returns

Table 3.3 summarises the descriptive statistics for the daily log-returns. Consistent with financial theory, the unconditional mean for both series is close to zero. However, the EEM exhibits a higher standard deviation (0.0173) compared to the DXY (0.0047), reflecting its higher risk profile.

Both series display deviations from normality. The Jarque-Bera test rejects the null hypothesis of a Gaussian distribution for both assets at the 0.01 significance level. While skewness is low for both series (0.035 for EEM and -0.081 for DXY), this non-normality is primarily driven by leptokurtosis. The kurtosis of the EEM reaches 18.96, reflecting the higher frequency of large price fluctuations and market shocks in developing economies.

Table 3.3: Descriptive statistics of daily log-returns (2005–2026)

Asset	Mean	Std. Dev.	Min	Max	Skewness	Kurtosis	JB Stat	<i>p</i> -value
EEM	0.0003	0.0173	-0.1763	0.2051	0.035	18.96	56500.6	< 0.001
DXY	0.0000	0.0047	-0.0271	0.0252	-0.081	5.07	952.4	< 0.001

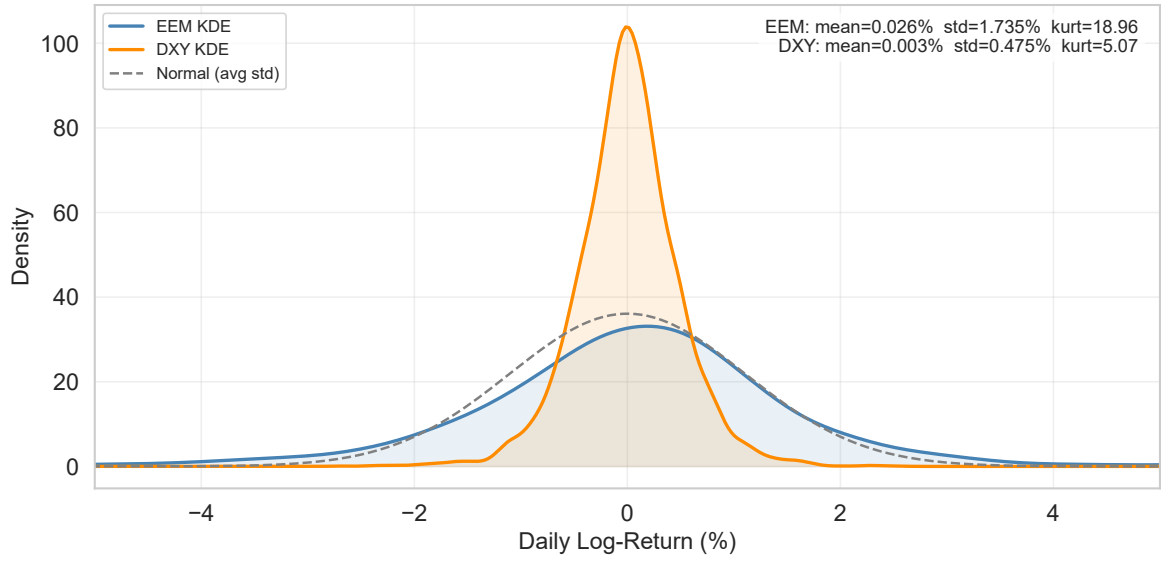
Note. The Jarque-Bera test evaluates the null hypothesis of a normal distribution. The strict rejection ($p < 0.01$) for both assets confirms significant departure from normality, driven primarily by excess kurtosis rather than skewness. A 5% significance level is used.

Source: author’s own calculations.

The difference in risk profiles is evident when examining the extreme values. The EEM recorded a single-day range of -17.63% to 20.51% during the 2008 Global Financial Crisis, whereas the DXY sample range spans from -2.71% to 2.52% . This reflects the higher vulnerability of the emerging market index to sudden capital flight and global shocks. Although non-normality does not invalidate the ARIMA framework – as OLS estimates remain consistent by the Law of Large Numbers and asymptotically normal by the Central Limit Theorem (Tsay, 2010) – it motivates the use of the Student’s t -distribution for the innovations in the GARCH specification.

As illustrated in Figures 3.3 and 3.4, both series exhibit a leptokurtic shape in the Kernel Density Estimation (KDE) curves, characterised by heavy tails. The DXY displays a sharper central peak, while the EEM shows a substantially larger departure from the Gaussian benchmark due to extreme tail events. The Q-Q plots confirm that Gaussian models underestimate tail risk in both series (Nelson, 1991; Tsay, 2010).

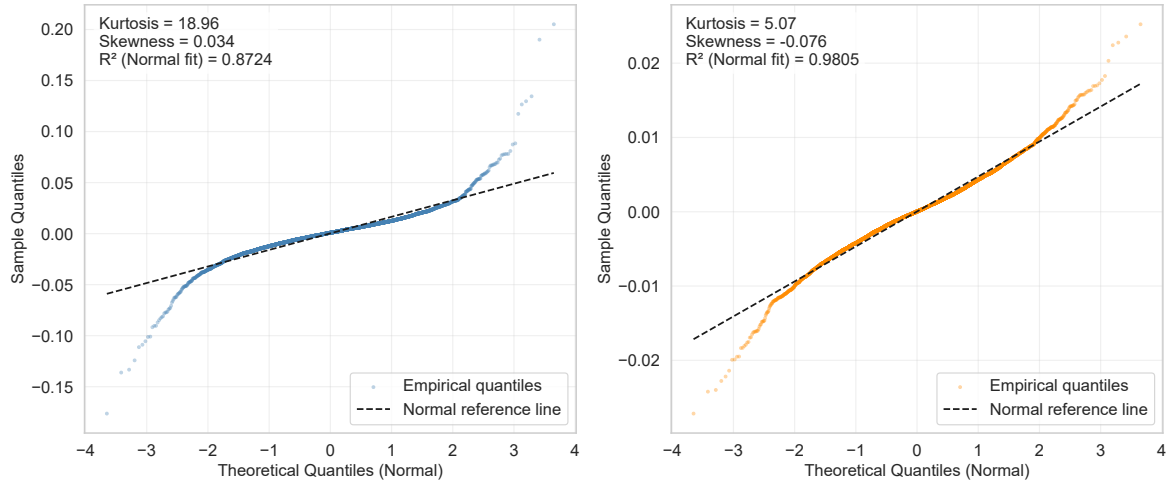
Figure 3.3: Return Distribution Overlay: EEM vs DXY against a Normal curve



Note. The dashed curve is a simplified Gaussian reference used for visual illustration only; it does not represent a fitted model for either series.

Source: author's own calculations.

Figure 3.4: Q-Q Plot vs Normal Distribution: EEM and DXY



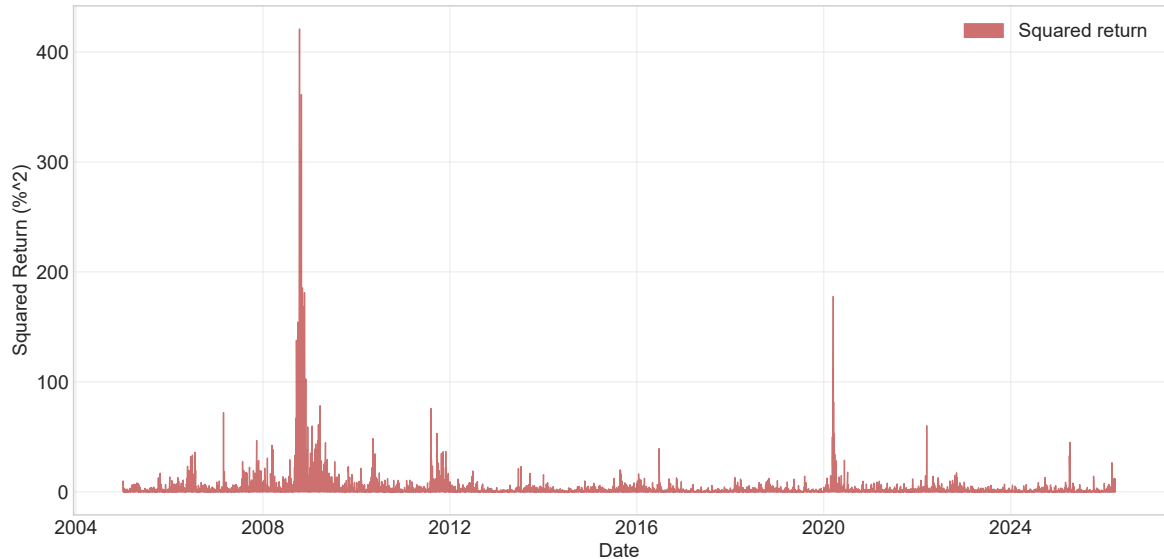
Source: author's own calculations.

3.4 Volatility Clustering

A stylised fact of empirical financial time series is the phenomenon of volatility clustering. Contrary to the homoskedasticity assumption of classical linear models, the variance of financial returns is rarely constant over time; periods of heightened volatility typically cluster together, as do periods of relative calm, so that large absolute returns tend to be followed by further large moves, while tranquil periods exhibit persistently small fluctuations (Mandelbrot, 1963; Tsay, 2010).

Figure 3.5 illustrates this phenomenon using the squared daily log-returns of EEM as a proxy for realised variance. The variance arrives in concentrated bursts rather than being uniformly distributed. Periods of low market activity are interspersed with pronounced volatility spikes, most notably during the 2008 Global Financial Crisis and the 2020 COVID-19 shock.

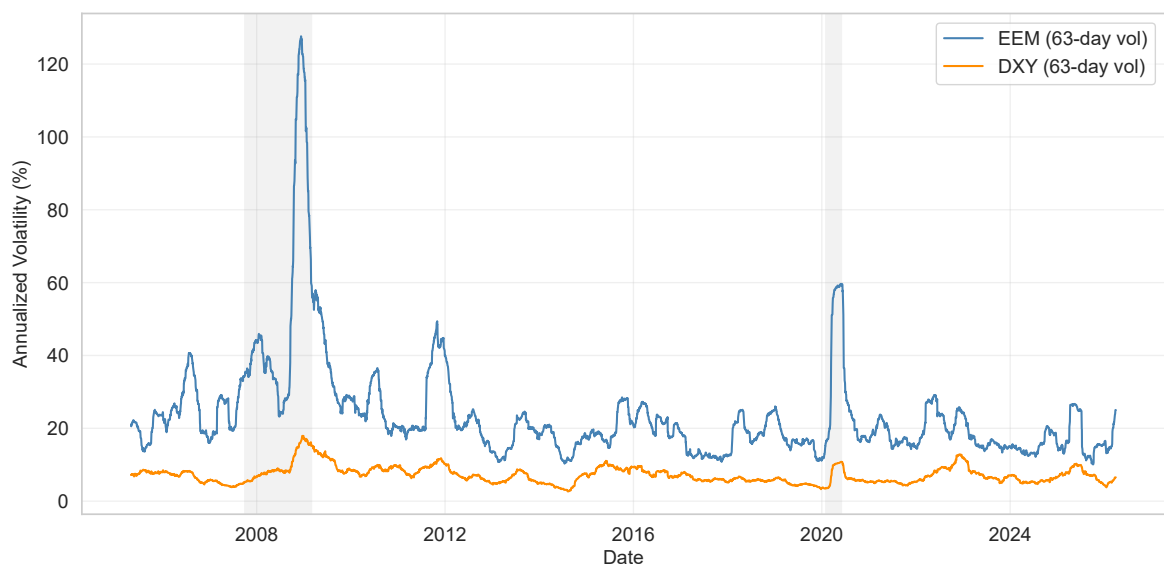
Figure 3.5: Squared Returns (Volatility Proxy) for EEM



Source: author's own calculations.

To further compare the risk profiles of these structurally distinct assets, Figure 3.6 illustrates their 63-day rolling annualised volatility. The chart highlights the highly regime-dependent nature of market risk, with volatility oscillating between periods of relative calm and extreme market stress.

Figure 3.6: Rolling 63-Day Annualised Volatility: EEM vs DXY



Source: author's own calculations.

While the EEM exhibits higher volatility than the DXY across all market regimes – reflecting its risk profile as an equity proxy compared to a reserve currency – their volatility spikes are synchronised. During global macroeconomic shocks, the variance of both assets increases concurrently.

To statistically evaluate this phenomenon of volatility clustering, the Lagrange multiplier test for autoregressive conditional heteroskedasticity (ARCH-LM) is applied to the daily return series (Tsay, 2010). The test results are summarised in Table 3.4.

Table 3.4: Tests for Conditional Heteroskedasticity (Volatility Clustering)

Asset	ARCH-LM Statistic	p -value	Null Hypothesis (H_0)	Conclusion
EEM	1656.77	< 0.001	Homoskedasticity	Rejected
DXY	300.51	< 0.001	Homoskedasticity	Rejected

Note. The ARCH-LM test was applied with $q = 5$ lags. The null hypothesis posits no autoregressive conditional heteroskedasticity. A p -value < 0.05 indicates rejection at the 5% significance level.

Source: author's own calculations.

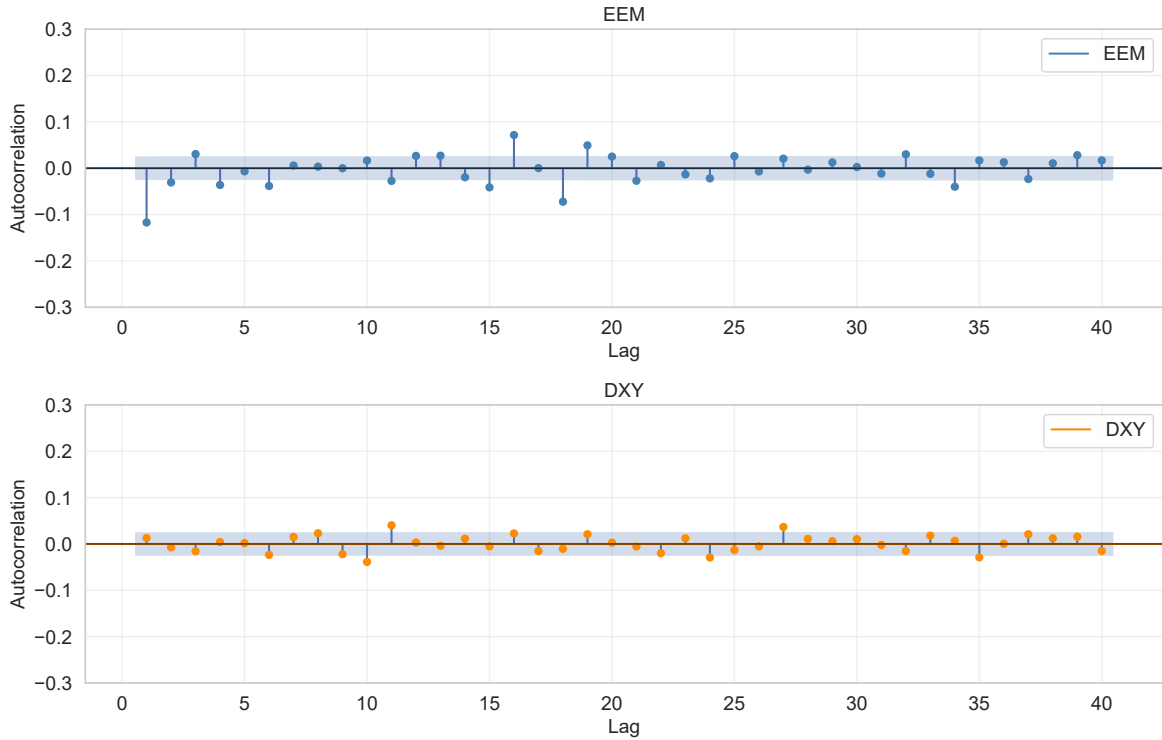
For both the emerging market proxy (EEM) and the fiat benchmark (DXY), the ARCH-LM test yields test statistics of 1656.77 and 300.51, respectively. Consequently, the null hypothesis of homoskedasticity is rejected at the 0.01 significance level. This confirms the presence of volatility clustering, justifying the application of the conditional heteroskedasticity framework (GARCH models) detailed in Section 2.6.

3.5 Autocorrelation Structure

To assess the presence of linear predictability and serial dependence within the data, the sample Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the return series are examined. As illustrated in Figure 3.7, the ACF of the raw daily log-returns exhibits minor spikes across 40 lags. However, relying on visual diagnostics of individual lags can be insufficient, as financial applications often require jointly testing whether several autocorrelations are zero (Tsay, 2010).

To statistically evaluate the joint significance of these autocorrelations, the Ljung-Box Q -test is applied. While individual t -tests assess single lags, the Ljung-Box Q -statistic tests the joint null hypothesis that the first m autocorrelations are simultaneously equal to zero. This provides a joint diagnostic procedure for detecting linear structure in asset returns. The test results are summarised in Table 3.5.

Figure 3.7: Autocorrelation Function (ACF) of daily log-returns: EEM vs DXY (40 lags)



Source: author's own calculations.

Table 3.5: Ljung-Box Test for Joint Serial Correlation (Daily Log-Returns)

Asset	Ljung-Box Q -Statistic	p -value	Conclusion
EEM	100.85	< 0.001	Autocorrelation Detected
DXY	22.20	0.0141	Autocorrelation Detected

Note. The Ljung-Box Q -statistic tests the joint null hypothesis of no serial correlation up to lag $m = 10$. A p -value < 0.05 indicates rejection of the null hypothesis at the 5% significance level.

Source: author's own calculations.

The empirical findings presented in Table 3.5 reveal that the null hypothesis of no serial correlation is rejected for both assets at the 0.05 significance level. For the emerging market equity proxy (EEM), the test yields a Q -statistic of 100.85 ($p < 0.01$), indicating the presence of linear serial dependence in the return series. For the US Dollar Index (DXY), the Q -statistic of 22.20 ($p = 0.0141$) similarly rejects the white-noise hypothesis.

The detected serial dependence has two direct implications for the modelling framework. First, it justifies applying the Box-Jenkins ARIMA framework, as the statistically significant autocorrelation indicates that the historical lag structure contains linear information that can be modelled explicitly. Second, it motivates the inclusion of lagged return features in the machine learning models, allowing non-linear algorithms to capture any residual dependencies that the linear ARIMA specification may not fully account for. Whether this autocorrelation structure constitutes a violation of the weak-form EMH is examined in Chapter 4.

3.6 Cross-Asset Predictors and Target Dependencies

To test the semi-strong form of the Efficient Market Hypothesis, the information set is expanded beyond historical prices to include publicly available macroeconomic data (Fama, 1991). Due to the structural differences between developing economies and the global reserve currency, two distinct feature spaces are constructed. Both predictive models utilise nine exogenous cross-asset instruments, with the specific composition reflecting the economic drivers of each target asset. In total, twelve macroeconomic indicators are employed across the two frameworks.

The selection of the cross-asset signals is based on established economic mechanisms. The developing markets proxy (EEM) is sensitive to global risk appetite and regional funding conditions. Therefore, its feature space incorporates the Volatility Index (VIX) as a gauge of market stress, alongside the Japanese Yen (JPY), which serves as a funding currency for emerging market investments. Conversely, the global reserve benchmark (DXY) is linked to monetary policy divergence between the United States and Europe. Consequently, its predictive model includes the Euro (EUR) and the British Pound (GBP), the dominant constituents of the DXY basket. Both frameworks share core global variables, including US Treasury yields to capture risk-free rate differentials, the S&P 500 index to proxy equity market returns, and industrial commodities (Crude Oil and Copper) to reflect the global business cycle.

Consistent with multifactor asset pricing theory (Chen et al., 1986), indicators such as Crude Oil and Copper capture global demand and inflationary pressures, while Gold reflects investor risk aversion. Incorporating these variables allows the ARIMAX and machine learning models to detect cross-asset spillovers. The transmission of these macroeconomic shocks is often non-linear and regime-dependent; cross-asset correlations increase during systemic crises, motivating the use of non-linear algorithms to capture these asymmetric effects. If the semi-strong Efficient Market Hypothesis holds, integrating these global signals should not significantly improve out-of-sample predictive accuracy over the baseline.

Table 3.6 summarises the selected exogenous variables, detailing their tickers, the target model to which they are applied, and the economic rationale for their inclusion. The complete descriptive statistics and formal stationarity test results for these predictors are provided in Appendix D.

Table 3.6: Summary of Exogenous Macroeconomic and Cross-Asset Variables

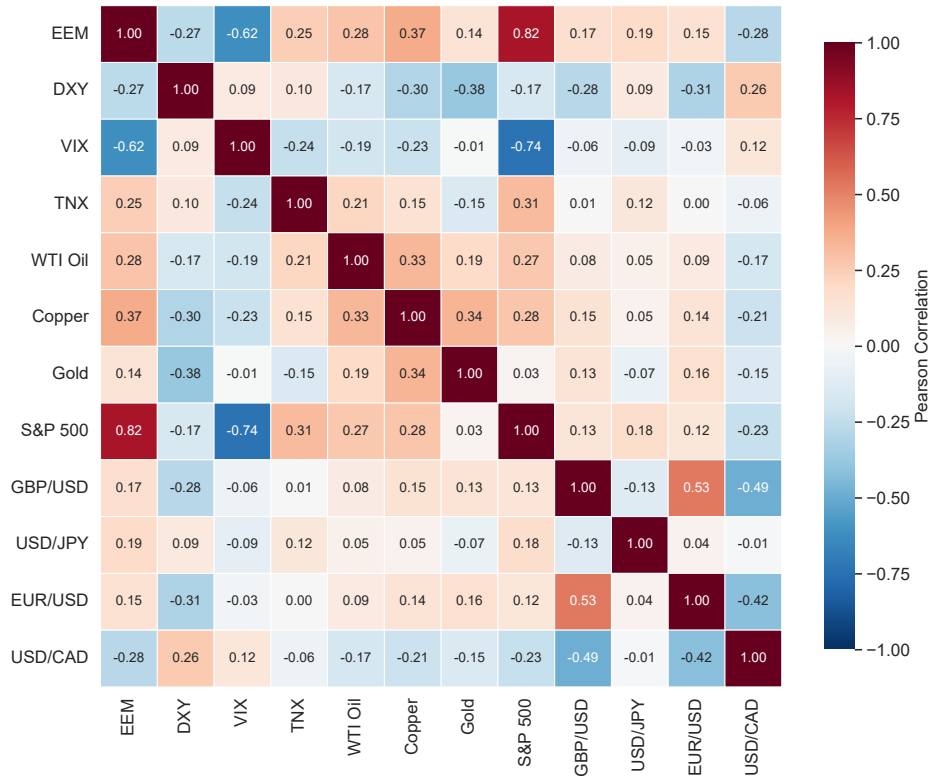
Variable	Ticker	Target	Economic Rationale
10Y US Treasury	$\hat{\text{TNX}}$	Both	Represents the global risk-free rate, dictating the cost of capital, debt-servicing burdens, and long-term yield differentials.
Crude Oil (WTI)	$\text{CL}=\text{F}$	Both	Proxy for the global energy cycle, acting as a crucial cost driver for developed markets and a revenue source for emerging economies.
Copper	$\text{HG}=\text{F}$	Both	A leading indicator of global industrial demand and manufacturing growth, reflecting the health of the global business cycle.
Gold	$\text{GC}=\text{F}$	Both	A traditional safe-haven asset, inflation hedge, and indicator of macroeconomic uncertainty and systemic risk.
S&P 500 Index	$\hat{\text{GSPC}}$	Both	Represents global equity sentiment and broad macroeconomic risk-on/risk-off business cycles.
USD / CAD	$\text{CAD}=\text{X}$	Both	A commodity-linked currency reflecting North American monetary policy differentials and serving as a proxy for global energy cycle dynamics.
US Dollar Index	$\text{DX}-\text{Y.NYB}$	EEM	Proxy for global liquidity constraint. A strong dollar typically signals capital outflows from emerging markets.
Volatility Index	$\hat{\text{VIX}}$	EEM	The primary gauge of global risk appetite. High VIX signals a “flight to quality” and systemic market stress, disproportionately affecting emerging market assets.
Japanese Yen	$\text{JPY}=\text{X}$	EEM	A primary global funding currency and a regional safe-haven proxy during periods of Asian and global market turbulence.
MSCI EM ETF	EEM	DXY	Captures aggregate emerging market equity performance; structurally inversely correlated with USD strength via global capital flow mechanics.
Euro / USD	$\text{EURUSD}=\text{X}$	DXY	The Euro is the largest constituent of the US Dollar Index basket, directly driving the valuation of the DXY.
British Pound / USD	$\text{GBPUSD}=\text{X}$	DXY	The British Pound is a major constituent of the DXY basket, reflecting European/US monetary policy divergence.

Source: All daily closing prices are sourced from Yahoo Finance (2026)

Figure 3.8 presents the Pearson correlation matrix encompassing both target indices and the exogenous variables. The heatmap illustrates contemporaneous co-movements that align with economic theory. Notably, the negative correlation between the MSCI Emerging Markets ETF (EEM) and the US Dollar Index (DXY) reflects the structural vulnerability of developing economies to a strong global reserve currency. Similarly, the positive correlations between the EEM, the S&P 500, and industrial commodities reflect the pro-cyclical nature of emerging market equities. Furthermore, the inverse relationship between the Volatility Index (VIX) and global equities illustrates flight-to-safety dynamics during market turbulence.

While these contemporaneous (lag-0) linkages justify the economic rationale for selecting these specific variables, they do not imply direct predictive power. To evaluate their true predictive capacity and prevent look-ahead bias, all exogenous variables are strictly lagged by one trading day ($t - 1$) prior to their inclusion in the forecasting models.

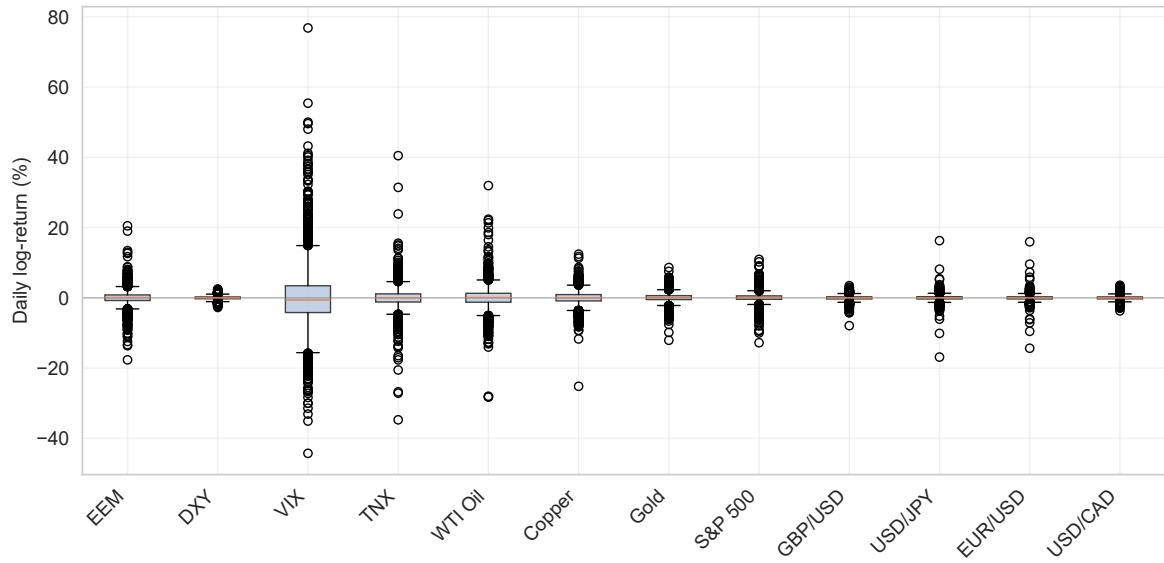
Figure 3.8: Full Pearson Correlation Matrix: Targets and Cross-Asset predictors



Source: author's own calculations.

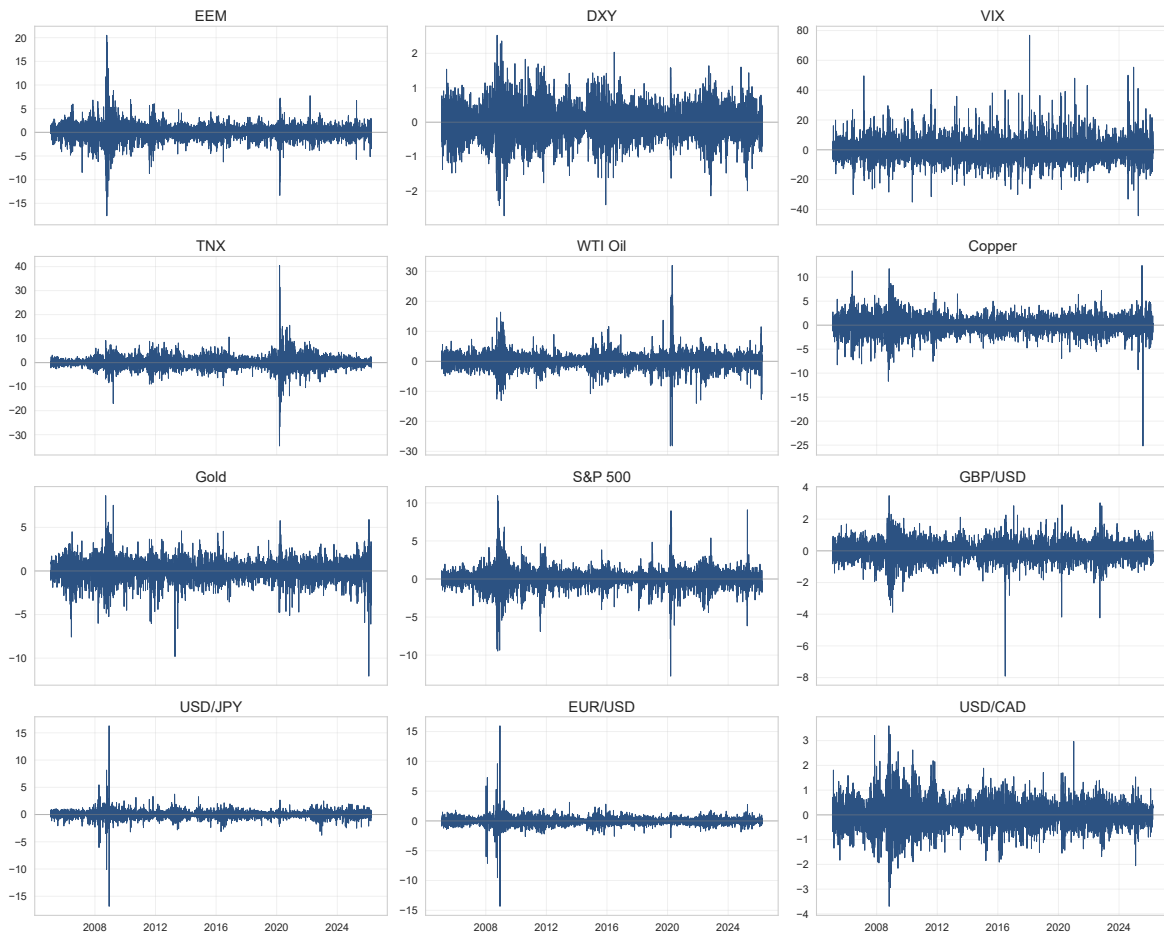
Having established the contemporaneous linkages between the variables, it is important to examine their individual empirical properties. The return distribution boxplots (Figure 3.9) and the temporal evolution of the cross-asset daily log-returns (Figure 3.10) illustrate the diverse volatility regimes within the feature space. Specifically, the pronounced positive skewness and large outliers observed in the VIX and energy markets contrast with the narrower, leptokurtic distributions of the fiat currency pairs. This structural heterogeneity indicates that each exogenous variable introduces a distinct risk profile into the modelling framework.

Figure 3.9: Return Distribution Boxplots: Comparative tail risk across all assets



Source: author's own calculations.

Figure 3.10: Cross-Asset Daily Log>Returns

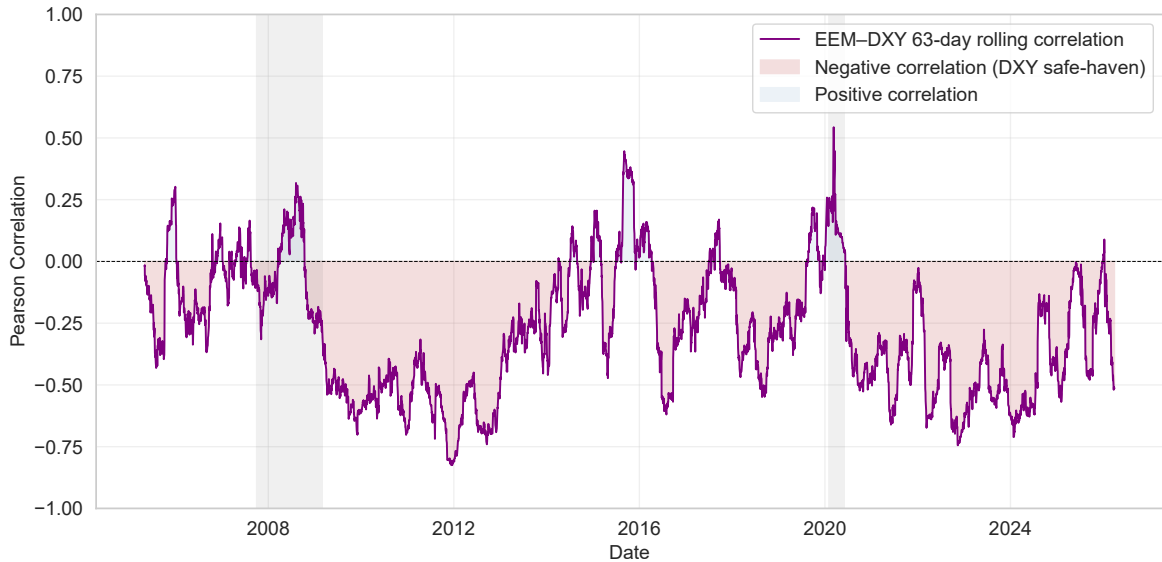


Source: author's own calculations.

The combination of cross-asset linkages and diverse volatility dynamics motivates the application of machine learning algorithms. Non-linear models can capture complex dependencies in noisy environments more effectively than traditional linear frameworks.

To further examine these dynamics, Figure 3.11 presents the 63-day rolling correlation between the two target assets, isolating the temporal evolution of their interdependence.

Figure 3.11: Rolling 63-Day Correlation: EEM vs DXY Returns



Source: author's own calculations.

Although the assets generally exhibit an inverse relationship, this negative correlation becomes more pronounced during periods of systemic stress – such as the 2008 Global Financial Crisis and the 2020 pandemic shock – when safe-haven demand for the US Dollar coincides with capital outflows from emerging market equities. Conversely, during periods of macroeconomic calm, the correlation frequently reverts toward zero. This temporal instability in the cross-asset linkage provides empirical justification for eschewing a static full-sample model fit. Instead, it motivates the dynamic, rolling-window out-of-sample forecasting architecture detailed in Section 2.5, allowing the predictive algorithms to adapt to shifting market regimes.

3.7 Final Dataset Specifications for Predictive Modelling

After examining the empirical data properties and macroeconomic linkages, the final preparation step involves constructing the input matrices required to test the Efficient Market Hypothesis (EMH). The information sets provided to the classical time series models and the machine learning algorithms differ to align with these theoretical definitions.

To evaluate the weak-form EMH, the classical Box-Jenkins framework – encompassing the constant-mean white-noise model (ARIMA(0,0,0), hereafter referred to as the naive baseline) and univariate ARIMA-GARCH models – relies on the univariate sequence of daily logarithmic

mic returns. In contrast, testing the semi-strong form of market efficiency requires expanding the information set to include publicly available macroeconomic data. For the econometric approach, this is achieved via the ARIMAX-GARCH specification, which incorporates the nine exogenous cross-asset returns detailed previously. To prevent look-ahead bias, all exogenous variables are lagged by one trading day (shifted to $t - 1$). Table 3.7 summarises the information sets provided to these models.

Table 3.7: Econometric Dataset Specifications for EMH Testing

EMH Form Tested	Model Framework	Input Variables	Dimensionality
Weak-form	ARIMA(0,0,0) & ARIMA-GARCH	Target returns	1
Semi-strong-form	ARIMAX-GARCH	Target returns + 9 Exog.	10

Note. The 9 exogenous variables correspond exactly to the macroeconomic predictors detailed in Section 3.6.

While the ARIMAX framework tests the semi-strong EMH using a concise set of linear regressors, standard machine learning algorithms lack built-in temporal memory. Unlike ARIMA-GARCH models, they do not natively incorporate autoregressive lag structures or endogenously model volatility clustering. Consequently, this temporal dimension must be engineered into the feature space. For both the emerging market proxy (EEM) and the fiat benchmark (DXY), the machine learning dataset comprises fifteen predictive features, constructed to replicate and extend the information available to the classical time-series framework.

This expanded matrix includes four autoregressive lags of the target variable (lags 1, 2, 3, and 5) to capture internal momentum without relying on a predefined ARMA structure. Furthermore, the nine macroeconomic cross-asset returns are included, alongside two risk proxies: the 21-day and 63-day rolling standard deviations of the target returns. Because standard rolling-window calculations inherently incorporate the observation at time t , these computed volatility metrics are subsequently lagged by one trading day (shifted to $t - 1$) to prevent look-ahead bias. This modification enables the non-linear algorithms to account for shifting regimes and volatility clustering while strictly preserving the chronological information set. Table 3.8 summarises this 15-dimensional feature space.

To evaluate out-of-sample performance, the dataset is chronologically partitioned into a training set (the first 80% of observations) and a hold-out test set (the remaining 20%). Because regularised regressions and distance-based algorithms such as SVR are highly sensitive to input scales, the features are standardised using z-score normalisation. To prevent data leakage, the standardisation parameters (mean and variance) are estimated exclusively on the training partition and subsequently applied to scale the test set.

Table 3.8: Feature Space Specifications for Machine Learning Models (15 variables)

Feature Category	DXY Model Inputs	EEM Model Inputs	Temporal Shift
Autoregressive Lags (4 variables)	DXY Returns (Lags 1, 2, 3, 5)	EEM Returns (Lags 1, 2, 3, 5)	None (inherent)
Exogenous Cross-Assets (9 variables)	EEM, Copper, 10Y Yield, Gold, Oil, EUR/USD, GBP/USD, USD/CAD, S&P 500	DXY, VIX, 10Y Yield, Oil, Copper, Gold, S&P 500, USD/CAD, JPY	Shifted to $t - 1$
Rolling Volatility (2 variables)	DXY 21-day Std. Dev. DXY 63-day Std. Dev.	EEM 21-day Std. Dev. EEM 63-day Std. Dev.	Shifted to $t - 1$

Note. All contemporaneous features and volatility proxies are strictly lagged by one trading day ($t - 1$) prior to model ingestion to prevent data leakage and guarantee that out-of-sample forecasts rely exclusively on historical information.

Table 3.9 details the chronological boundaries and sample sizes of the data partitions. Due to the 63-day warm-up period required for computing the rolling volatility features, the effective modelling sample for both the econometric and machine learning models begins in May 2005. This ensures an identical evaluation horizon across all model classes.

Table 3.9: Chronological Dataset Partitioning (80/20 Split)

Target Asset	Partition	Start Date	End Date	Observations
US Dollar Index (DXY)	In-Sample (Train)	2005-05-02	2022-01-19	4,210
	Out-of-Sample (Test)	2022-01-20	2026-03-30	1,053
Emerging Markets (EEM)	In-Sample (Train)	2005-05-03	2022-01-18	4,206
	Out-of-Sample (Test)	2022-01-19	2026-03-30	1,052

Note. The partitioning strictly maintains the chronological order of the time series to prevent look-ahead bias.

With the exploratory data analysis complete and the feature matrices constructed to test both forms of market efficiency, the analysis proceeds to Chapter 4, which presents the empirical results of the out-of-sample forecasting and the statistical tests of the Efficient Market Hypothesis.

4. Empirical Results

This chapter presents the empirical findings from the out-of-sample forecasting exercise, aiming to evaluate the predictive capabilities of classical econometric methods and machine learning algorithms.

The analysis is structured around the standard classifications of the Efficient Market Hypothesis (EMH). Section 4.1 investigates the weak-form EMH using only the historical price sequences of the target assets. It explores their statistical properties via Variance Ratio and Runs tests, and evaluates univariate ARIMA-GARCH models to assess return predictability and conditional volatility structure.

Section 4.2 shifts focus to the semi-strong form EMH by incorporating publicly available macroeconomic indicators and cross-asset returns. The framework is expanded into multivariate ARIMAX specifications, regularised regressions (Ridge, Lasso), and non-linear machine learning algorithms (Random Forest, XGBoost, SVR) to determine whether they can extract out-of-sample predictive signals.

Section 4.3 addresses the three research hypotheses, evaluating whether the collective forecasting results support or reject market efficiency for DXY and EEM, and discussing their implications for return predictability and market risk.

Finally, all empirical estimations and data processing were conducted in Python. The research pipeline utilised *yfinance* (Aroussi, 2020) for downloading historical market data, alongside *pandas* (McKinney, 2010) and *NumPy* (Harris et al., 2020) for data manipulation and numerical calculations. The classical time-series econometrics – comprising ARIMA/ARIMAX and GARCH/ARX-GARCH frameworks – were implemented using *statsmodels* (Seabold & Perktold, 2010) and *arch* (Sheppard et al., 2024), with *pmdarima* handling the automated lag selection (Smith et al., 2017). Machine learning algorithms and regularised regressions (Ridge, Lasso, Random Forest, SVR, and Gradient Boosting) were deployed via *scikit-learn* (Pedregosa et al., 2011) and *xgboost* (Chen & Guestrin, 2016). Furthermore, *SciPy* (Virtanen et al., 2020) was used to conduct formal statistical tests and compute *p*-values, while *Matplotlib* (Hunter, 2007) and *Seaborn* (Waskom, 2021) were employed for all graphical visualisations. The implementation is provided in Appendix A.

4.1 Evaluating Weak-Form Market Efficiency

4.1.1 Statistical Tests for a Random Walk

As noted by Fama (1991), a distinction exists between statistical predictability and economic inefficiency. While large financial datasets may exhibit in-sample statistical dependencies,

these do not necessarily imply exploitable economic opportunities. Before deploying the forecasting models, the return series are analysed using the Variance Ratio and Runs tests, as described in Section 2.4. These tests provide a model-independent assessment of linear and non-linear serial dependence within the data.

Table 4.1 presents the results of the Variance Ratio test, and Table 4.2 summarises the Runs test statistics for both DXY and EEM. These tests serve as the initial step in evaluating the weak-form Efficient Market Hypothesis.

Table 4.1: Variance Ratio Test (Lo-MacKinlay)

Target Asset	Lag (k)	Variance Ratio	p -value
DXY	2	1.0118	0.4919
	5	0.9987	0.9718
	10	0.9900	0.8630
EEM	2	0.8823	< 0.001
	5	0.7847	0.0048
	10	0.7057	0.0138

Note. The null hypothesis of a random walk (Variance Ratio = 1) is tested using the heteroskedasticity-robust statistic of Lo and MacKinlay (1988). p -values are two-sided ($\alpha = 0.05$).

Source: author's own calculations.

Table 4.2: Runs Test (Wald-Wolfowitz) for Directional Randomness

Target Asset	Observed Runs	Expected Runs	Z -statistic	p -value
DXY	2703	2632.40	1.9467	0.0516
EEM	2701	2622.65	2.1673	0.0302

Note. The Runs test evaluates whether the sequence of positive and negative returns is consistent with the null hypothesis of directional randomness. p -values are two-sided ($\alpha = 0.05$).

Source: author's own calculations.

The in-sample empirical findings reveal a structural divergence between the two asset classes. For DXY, the statistics are consistent with the weak-form Efficient Market Hypothesis. Both the Variance Ratio test (yielding ratios statistically indistinguishable from unity) and the non-parametric Runs test ($Z = 1.9467$, $p = 0.0516$) fail to reject the random walk null hypothesis at the 5% significance level, indicating no significant linear or directional dependencies.

Conversely, the statistical profile of EEM exhibits clear departures from a random walk. Its Variance Ratio drops to 0.7057 at lag 10, indicating a tendency toward mean-reverting behaviour over longer horizons. Similarly, the Runs test ($Z = 2.1673$, $p = 0.0302$) detects structural dependencies in the sequence of positive and negative returns.

This divergence suggests that while the developed foreign exchange market proxy (DXY) is

consistent with weak-form efficiency in-sample, the emerging markets equity proxy (EEM) retains a measurable degree of historical memory. These results directly inform the univariate model specifications detailed in Section 4.1.2.

4.1.2 Evaluating Univariate Return Predictability

Transitioning to the predictive evaluation of the weak-form EMH, the baseline representing an efficient market is defined as the ARIMA(0,0,0) specification (Section 2.6). To test whether historical memory can outperform this baseline, two frameworks are employed: a Two-Step ARIMA(p,d,q)+GARCH(1,1) and a Joint Maximum Likelihood AR(p)+GARCH(1,1) model.

This dual approach addresses algorithmic constraints within the Python environment, which lacks joint Maximum Likelihood Estimation support for models with Moving Average terms. Thus, extracting the full ARIMA(p,d,q) structure requires a two-step procedure. Following Tsay (2010), this serves as a practical approximation, although its exact statistical properties are not rigorously established. Therefore, a Joint AR(p)+GARCH(1,1) model is evaluated alongside it. By restricting the mean equation to autoregressive terms, both equations are estimated simultaneously using a Student-*t* distribution to obtain exact standard errors.

In-Sample Model Selection and Diagnostics

Before deploying the forecasting models out-of-sample, their optimal lag structures were identified on the training data by minimising the Akaike Information Criterion (AIC). The two-step framework utilised an automated search via the *pmdarima* library (Smith et al., 2017), while the AR+GARCH framework employed a discrete grid search over a predefined lag space. For DXY, no specification yielded a lower AIC than the zero-lag model, resulting in an ARIMA(0,0,0) and an AR(0) structure. This aligns with the in-sample weak-form efficiency of the mature currency market.

Conversely, the EEM proxy exhibited historical dependence. The automated search identified an ARIMA(2,0,1) as the optimal two-step structure. For the joint model, Table 4.3 tracks the AIC values across different lag orders. Although the AIC continued to decrease marginally beyond the fifth lag, the relative reduction diminished. To balance explanatory power with parametric parsimony, the AR(5) specification was selected.

These algorithmic selections corroborate the exploratory data analysis in Section 3.5. As established by the formal Ljung-Box statistics, the emerging market proxy contains significantly more serial correlation than the US Dollar Index. The AIC minimisation captures these preliminary diagnostics, incorporating the detected linear dependencies into the selected model specifications.

Table 4.3: AIC Lag Selection for AR(p)+GARCH(1,1) (In-Sample)

Lag Order (p)	DXY AIC	EEM AIC
$p = 0$	4950.05	14773.53
$p = 1$	4951.45	14761.20
$p = 2$	4952.74	14755.29
$p = 3$	4951.25	14754.13
$p = 4$	4951.90	14750.97
$p = 5$	4950.69	14746.51

Note. Minimum AIC values determining the final model order are highlighted in bold.

Source: author's own calculations.

To assess the significance of the selected lag structures and the inclusion of the conditional variance component, Likelihood Ratio Tests (LRT) were applied to both estimation frameworks, following the methodology described in Section 2.3. The results are summarised in Table 4.4.

Table 4.4: Likelihood Ratio Tests for In-Sample Model Validation

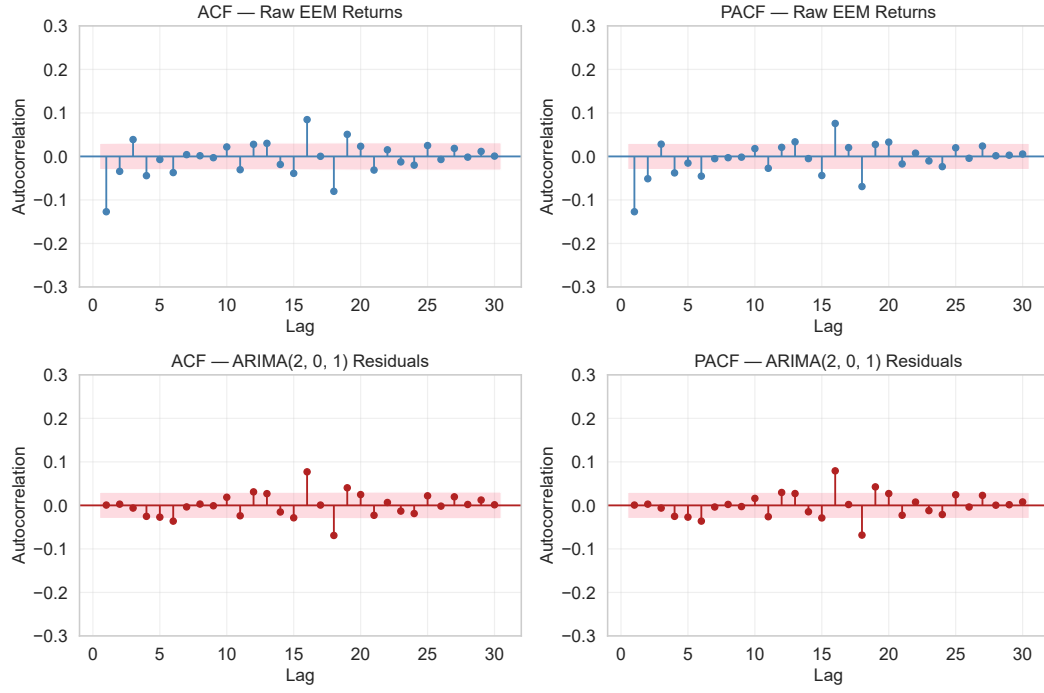
Target Asset	Tested Structure	Two-Step χ^2	Two-Step p -value	Joint MLE χ^2	Joint MLE p -value
DXY	GARCH(1,1) vs Constant Var.	471.95	< 0.001	471.96	< 0.001
EEM	AR/MA vs Constant Mean	81.31	< 0.001	37.02	< 0.001
	GARCH(1,1) vs Constant Var.	859.83	< 0.001	885.03	< 0.001

Note. For the Two-Step framework, LRT p -values for the mean equation are approximate, as they are based on the first-stage Gaussian log-likelihood and ignore the subsequent GARCH fit. The Joint MLE provides exact simultaneous test statistics.

Source: author's own calculations.

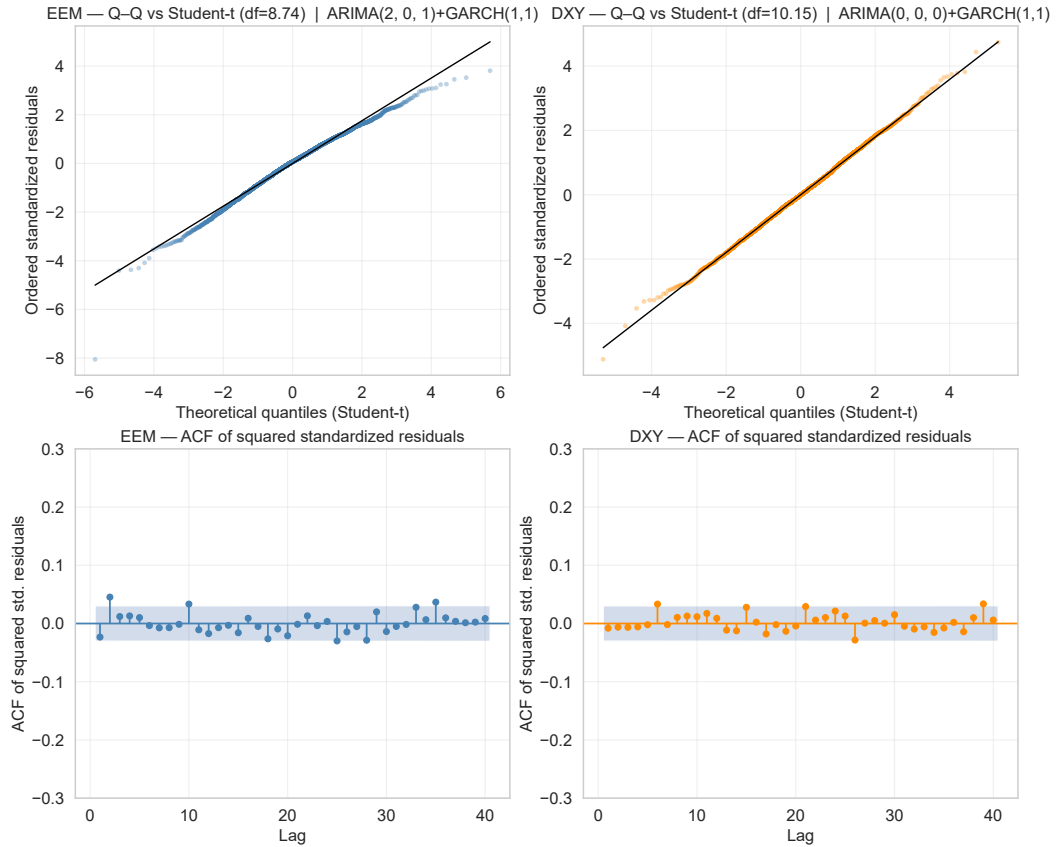
The χ^2 statistics derived from comparing the selected EEM mean equations against the naive baseline rejected the null hypothesis across both frameworks, indicating linear dependence in the historical price sequences. The same Likelihood Ratio Test was inapplicable to the DXY. Because the model selection identified the ARIMA(0,0,0) and AR(0) specifications as optimal, there were no additional parameters to evaluate against the constant-mean baseline. The inclusion of the GARCH(1,1) variance equation was supported for both assets, yielding p -values near zero. Figure 4.1 illustrates the ACF and PACF of raw returns versus ARIMA residuals, showing that the selected orders accounted for the linear dependence. Finally, Figure 4.2 presents the residual diagnostics used to assess the adequacy of the fitted GARCH(1,1) variance structures.

Figure 4.1: ACF and PACF: Raw EEM Returns vs ARIMA Residuals



Source: author's own calculations.

Figure 4.2: ARIMA + GARCH(1,1) Residual Diagnostics



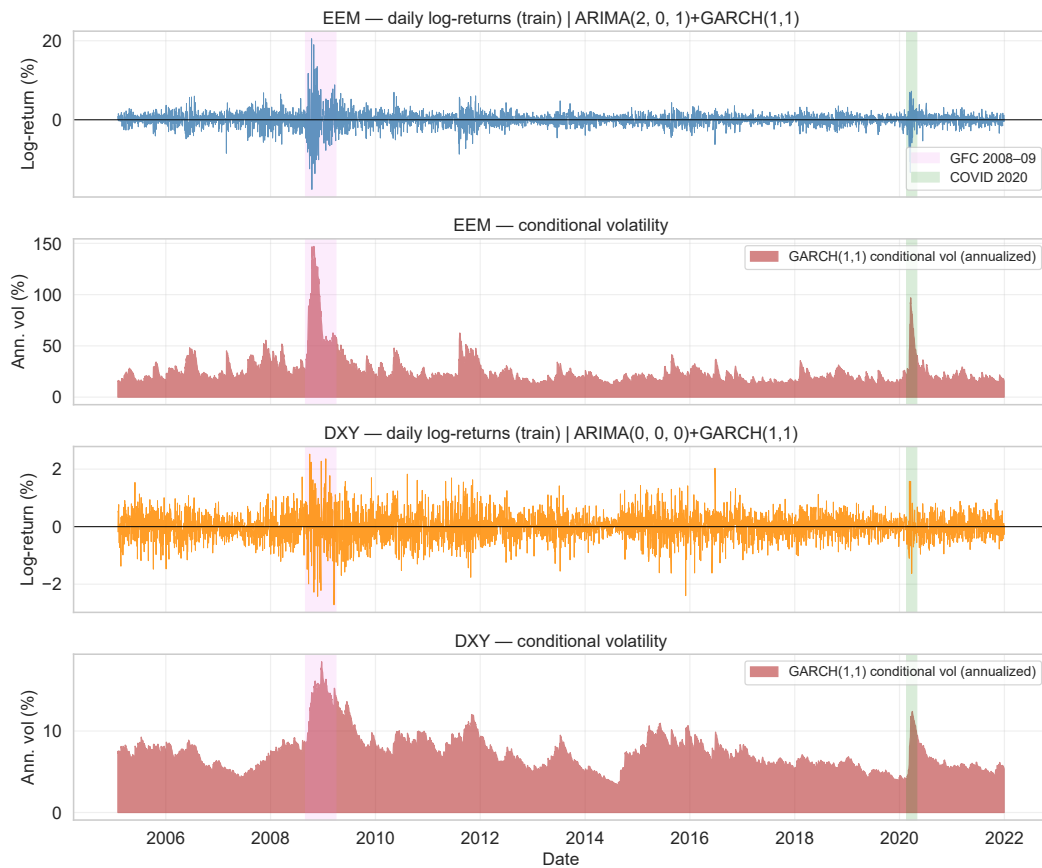
Source: author's own calculations.

The ARIMA(2,0,1) specification accounted for the minor autocorrelations in the raw EEM series, with residual autocorrelations remaining within the 95% confidence bounds (Figure 4.1). The diagnostic panels (Figure 4.2) show that the standardised residuals align with the Student- t quantiles, with left-tail deviations for EEM indicating that the distribution does not fully capture extreme negative returns. Furthermore, the squared residual correlograms exhibit no significant spikes, indicating that the GARCH(1,1) equation accounted for the conditional heteroskedasticity in both assets.

Volatility Modelling

Figure 4.3 presents the raw daily log-returns alongside the estimated annualised conditional volatility from the GARCH(1,1) specification for both the DXY and EEM indices. The daily conditional standard deviation is annualised by multiplying it by $\sqrt{252}$ to allow for a more intuitive economic interpretation. The plots indicate that the model accounts for the volatility clustering observed in the empirical data. Specifically, the estimated conditional volatility increases during known market stress events – including the 2008 Global Financial Crisis and the 2020 COVID-19 shock – and decreases during stable market conditions. This indicates that a constant-variance assumption would be inadequate for these financial series.

Figure 4.3: Two-step ARIMA + GARCH(1,1) – returns and conditional volatility



Source: author's own calculations.

Because a proper evaluation relies on formal statistical testing rather than solely on visual inspection, two diagnostic tests were applied, following the methodology detailed in Section 2.3, to assess whether the selected GARCH(1,1) specifications accounted for the conditional heteroskedasticity: the Ljung-Box Q^2 -test and the ARCH-LM test. An adequate volatility model yields standardised residuals with no significant serial correlation in their squared terms. Table 4.5 reports the p -values of these tests for the raw returns (initially evaluated in Sections 3.4 and 3.5) and the GARCH(1,1) standardised residuals.

Table 4.5: Residual Diagnostics for Conditional Heteroskedasticity

Target Asset	Evaluated Series	Ljung-Box Q^2 p -value	ARCH-LM p -value
DXY	Raw Returns	< 0.001	< 0.001
	GARCH(1,1) Standardised Residuals	0.7052	0.9783
EEM	Raw Returns	< 0.001	< 0.001
	GARCH(1,1) Standardised Residuals	0.0564	0.0259

Note. The Ljung-Box Q^2 test evaluates the joint null hypothesis of no serial correlation in the squared series up to lag $m = 20$; a p -value below 0.05 indicates rejection at the 5% significance level. The ARCH-LM test is implemented with $q = 5$ lags and tests the null hypothesis of no autoregressive conditional heteroskedasticity; again, a p -value below 0.05 implies rejection at the 5% level.

Source: author's own calculations.

The diagnostic results indicate that the selected models accounted for the conditional variance dynamics. For the DXY index, the GARCH(1,1) specification yielded p -values of 0.7052 and 0.9783, failing to reject the null hypothesis of no remaining ARCH effects. For the EEM proxy, the Ljung-Box test on the squared standardised residuals yields a p -value of 0.0564, indicating no significant autocorrelation at the 5% level. Although the ARCH-LM test produces a p -value of 0.0259 – rejecting the null at the 5% level but not at the 1% level – the substantial decrease in the test statistic compared to the raw returns indicates that the GARCH(1,1) equation accounted for the majority of the conditional heteroskedasticity.

The in-sample diagnostics indicate that the selected univariate models accounted for the linear dependence and volatility clustering in the data. The estimated parameters for all GARCH specifications, confirming high volatility persistence and heavy-tailed innovations, are detailed in Appendix B. The subsequent out-of-sample evaluation tests the weak-form EMH by assessing whether the modelled lag structures provide predictive power on unseen data.

Out-of-Sample Forecasting Performance

Following the in-sample analysis, the models were evaluated using a pseudo-out-of-sample expanding window approach. At each step, the training sample consists of T observations to generate a one-step-ahead forecast for $T + 1$, with the sample expanding chronologically and parameters re-estimated every 21 trading days. Following the methodology detailed

in Section 2.10, forecast accuracy was assessed against the ARIMA(0,0,0) baseline using the modified Diebold-Mariano (DM) test for squared losses, alongside a Binomial test for Directional Accuracy (DA). Table 4.6 reports these out-of-sample metrics.

Table 4.6: Out-of-Sample Performance of Univariate Models

Target Asset	Model Specification	RMSE	Dir. Accuracy	DM p -value	Binomial p -value
DXY	ARIMA(0,0,0)	0.004706	51.57%	–	0.3241
	AR(1)+GARCH(1,1) [†]	0.004702	49.67%	0.3541	0.8533
EEM	ARIMA(0,0,0)	0.011834	51.62%	–	0.3089
	ARIMA(2,0,1)+GARCH(1,1)	0.011870	48.10%	0.3680	0.2292
	AR(5)+GARCH(1,1)	0.011870	51.90%	0.1421	0.3392

Note. The Diebold-Mariano (DM) test compares the squared forecast errors of the specified model against the naive baseline. [†]The AR(1)+GARCH(1,1) specification for DXY is included for illustrative purposes only; the AIC-selected optimal model is the zero-lag AR(0)+GARCH(1,1).

Source: author's own calculations.

Based on the results of the estimated models, the out-of-sample findings indicate that neither asset's return series is predictable from its own historical sequence alone. For the DXY index, no exploitable linear patterns were detected during the in-sample estimation phase. The application of an AR(1) specification yielded a modest root mean square error (RMSE) reduction that is not statistically significant (p -value = 0.3541) according to the Diebold-Mariano test. Furthermore, the directional accuracy of 49.67% is consistent with random guessing (p -value = 0.8533). The out-of-sample performance of the DXY index is therefore consistent with a random walk.

For the EEM proxy, the out-of-sample evaluation contrasts with the in-sample findings. Although the Variance Ratio and Wald-Wolfowitz Runs tests (Tables 4.1 and 4.2) indicated significant internal memory and directional dependence within the training data, the out-of-sample application of the models did not improve upon the baseline. Both models yielded an RMSE of 0.011870, compared to the 0.011834 baseline. The DM test indicates that this difference is not statistically significant (p -values of 0.3680 and 0.1421 for the ARIMA(2,0,1) and AR(5) specifications, respectively), demonstrating that the modelled internal lag structure provides no predictive power on unseen data.

The out-of-sample evaluation indicates that directional returns exhibit no predictability based on their own historical sequence. Nevertheless, the in-sample diagnostics demonstrate that the observed volatility clustering can be modelled. As reported in Table 4.5, the GARCH(1,1) specification accounted for the conditional heteroskedasticity, leaving no significant ARCH effects in the standardised residuals. This establishes the baseline for Section 4.2, which investigates whether incorporating exogenous macroeconomic variables provides out-of-sample predictive power.

4.2 Evaluating Semi-Strong-Form Market Efficiency

Following the evaluation of weak-form efficiency, the analysis incorporates an expanded information set to test the semi-strong form of the Efficient Market Hypothesis (EMH). This section examines whether publicly available macroeconomic data and cross-asset returns provide a statistically significant improvement in out-of-sample forecasting accuracy. The methodology employs extended econometric specifications, regularised regressions, and non-linear machine learning algorithms. To prevent look-ahead bias, all exogenous variables are lagged by one trading day ($t - 1$).

4.2.1 Two-Step ARIMAX and ARX-GARCH Frameworks

To evaluate the semi-strong form of the Efficient Market Hypothesis, the univariate models were extended into two-step ARIMAX and ARX-GARCH specifications incorporating the nine cross-asset predictors. Re-estimating the optimal lag orders using the methodology detailed in Section 4.1.2 revealed that the lag structure for the EEM index remained unchanged, resulting in ARIMAX(2,0,1) and ARX(5) specifications. Conversely, the DXY specifications shifted to ARIMAX(0,0,1) and ARX(5). The complete AIC rankings underlying this ARX-GARCH lag selection are provided in Appendix B. The emergence of the MA(1) term indicates that a first-order serial dependence persists in the residuals, as the cross-asset predictors do not fully exhaust the short-term autocorrelation of the DXY series.

Table 4.7 reports the macroeconomic variables identified as statistically significant. For completeness, the full sets of coefficient estimates for both models are detailed in Appendix C. These predictive coefficients should be interpreted in contrast to the contemporaneous relationships established in Section 3.6. Because the forecasting framework measures the lagged influence of t on $t + 1$, the signs of the estimated coefficients can diverge significantly from contemporaneous correlations. For instance, while the S&P 500 index exhibits a strong contemporaneous correlation with EEM (+0.82), its lagged predictive coefficient in the ARX model is negative (-0.2541). This sign reversal indicates that the predictive relationship captures a short-term adjustment or partial reversal dynamic on the subsequent trading day.

Table 4.7: Statistically Significant Exogenous Predictors in ARX(5)+GARCH(1,1)

Target Asset	Exogenous Variable	Coefficient	Std. Error	t -stat.	p -value
DXY	10Y Treasury ($\hat{\text{TNX}}$)	0.0106	0.0038	2.819	0.0048
	S&P 500 ($\hat{\text{GSPC}}$)	-0.0321	0.0124	-2.582	0.0098
EEM	CBOE Volatility ($\hat{\text{VIX}}$)	-0.0105	0.0045	-2.343	0.0191
	S&P 500 ($\hat{\text{GSPC}}$)	-0.2541	0.0464	-5.477	< 0.001

Note. Only variables significant at the 5% level ($p < 0.05$) are reported. Coefficients are derived from the Joint Maximum Likelihood estimation.

Source: author's own calculations.

Out-of-Sample Forecasting Performance

The optimally specified ARIMAX-GARCH and ARX-GARCH models (selected by minimising the AIC for optimal lag structure) were evaluated out-of-sample using the expanding window approach, with parameters re-estimated every 21 trading days. Table 4.8 reports their predictive accuracy against the ARIMA(0,0,0) baseline.

Table 4.8: Out-of-Sample Performance: Two-Step ARIMAX and ARX

Target Asset	Model Specification	RMSE	Dir. Accuracy	DM p -value	Binomial p -value
DXY	ARIMA(0,0,0)	0.004706	51.57%	–	0.3241
	ARIMAX(0,0,1)+GARCH	0.004679	50.62%	0.3334	0.7116
	ARX(5)+GARCH	0.004678	51.57%	0.3098	0.3241
EEM	ARIMA(0,0,0)	0.011834	51.62%	–	0.3089
	ARIMAX(2,0,1)+GARCH	0.012247	50.67%	0.0063*	0.6886
	ARX(5)+GARCH	0.012011	51.52%	0.0412*	0.1696

Note. * indicates that the model performed significantly *worse* than the baseline (DM statistic > 0 and p -value < 0.05). To evaluate the semi-strong EMH against the full information set, all nine exogenous variables are retained in these specifications; feature selection is subsequently addressed using Ridge and Lasso regressions in Section 4.2.2.

Source: author's own calculations.

Although specific cross-asset predictors demonstrated in-sample significance, incorporating these macroeconomic variables did not improve out-of-sample forecasting accuracy. For the DXY index, the Diebold-Mariano tests indicate that the ARIMAX(0,0,1)+GARCH(1,1) and ARX(5)+GARCH(1,1) specifications did not significantly outperform the naive baseline (p -values of 0.3334 and 0.3098, respectively).

For the EEM proxy, the addition of macroeconomic factors degraded forecasting performance. The ARX(5)+GARCH(1,1) specification yielded an out-of-sample RMSE of 0.012011, which the DM test confirms is statistically significantly worse than the naive baseline (p -value = 0.0412). Furthermore, Binomial tests revealed that the directional accuracy for the multivariate models across both assets remained statistically indistinguishable from a random outcome.

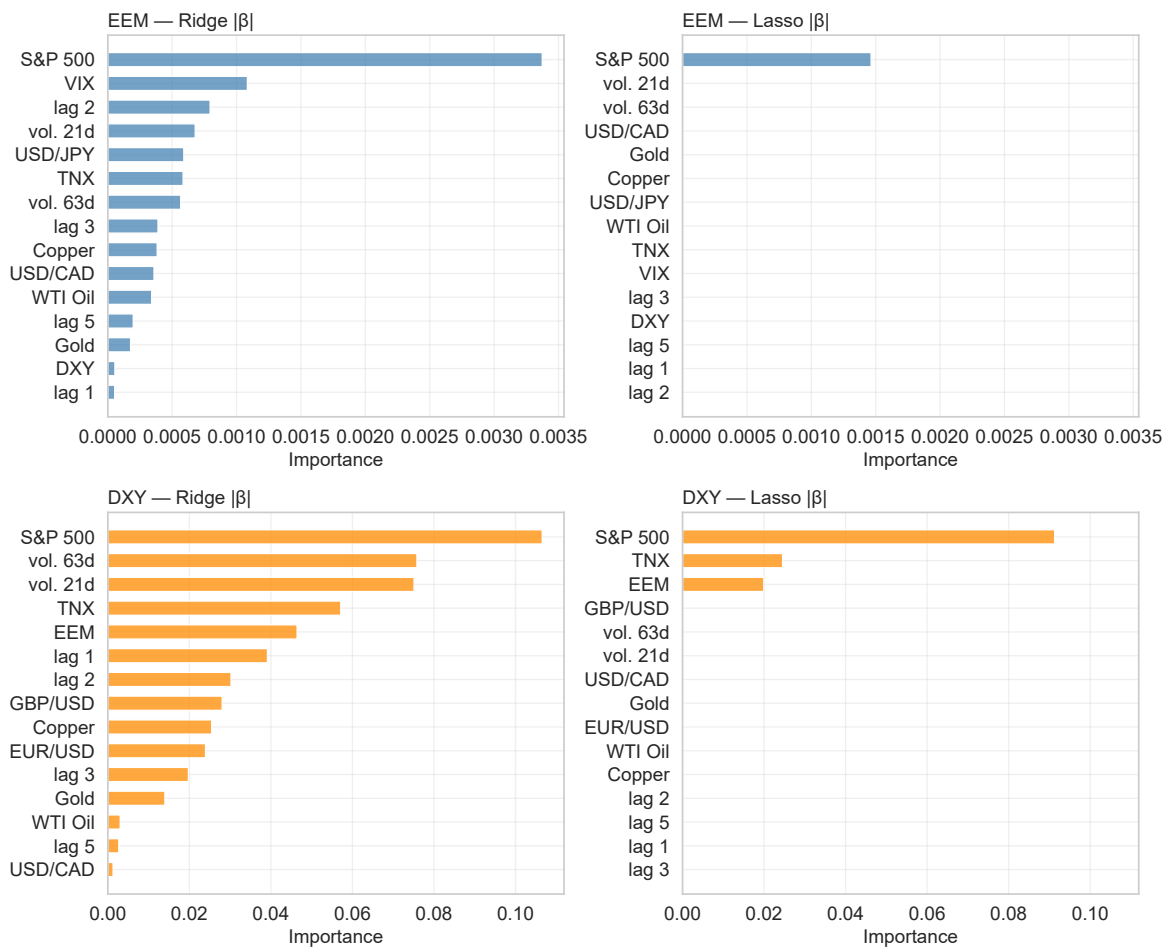
4.2.2 Regularised Linear Regression: Ridge and Lasso

While the classical ARIMAX framework incorporates macroeconomic features, standard unpenalised regressions are highly susceptible to multicollinearity and overfitting. To address this, regularised linear models – specifically Ridge (L_2 penalty) and Lasso (L_1 penalty) – are implemented. These models utilise an expanded 15-dimensional feature space, incorporating the 9 cross-asset returns, 4 autoregressive target lags, and 2 rolling volatility proxies.

In-Sample Feature Selection

To prevent look-ahead bias, the regularisation parameter (α) was selected via 5-fold Time-SeriesSplit cross-validation on the training data. The models were evaluated across 100-point logarithmic grids: $[10^{-3}, 10^2]$ for the Lasso regression and a wider range of $[10^{-3}, 10^5]$ for the Ridge regression. For the Ridge specification, cross-validation yielded penalty values of $\alpha \approx 657.9$ for the DXY index and $\alpha \approx 7390.7$ for the EEM. Conversely, the L_1 penalty of the Lasso regression performed embedded feature selection, reducing the feature space to four active predictors for the DXY ($\alpha \approx 0.0231$) and one active predictor for the EEM ($\alpha \approx 0.0013$). The retained coefficients are plotted in Figure 4.4.

Figure 4.4: Feature Weights: Ridge vs Lasso Regression



For the DXY index, the Lasso regression retained the EEM proxy, 10-Year US Treasury yields, the GBP/USD exchange rate, and the S&P 500 index. For the EEM proxy, the S&P 500 index is the sole active predictor. The common retention of the S&P 500 index suggests a potential sensitivity of both asset classes to US equity market dynamics.

Out-of-Sample Forecasting Performance

To assess whether parameter shrinkage and embedded feature selection translate into out-of-sample predictive power, Table 4.9 reports the forecasting accuracy of the regularised models against the naive baseline.

Table 4.9: Out-of-Sample Performance: Ridge and Lasso Regressions

Target Asset	Model Specification	RMSE	Dir. Accuracy	DM p -value	Binomial p -value
DXY	ARIMA(0,0,0)	0.004706	51.57%	–	0.3241
	Ridge ($\alpha \approx 657.9$)	0.004681	51.66%	0.2712	0.2947
	Lasso ($\alpha \approx 0.0231$)	0.004674	52.23%	0.1328	0.1563
EEM	ARIMA(0,0,0)	0.011834	51.62%	–	0.3089
	Ridge ($\alpha \approx 7390.7$)	0.011910	49.81%	0.2038	0.9263
	Lasso ($\alpha \approx 0.0013$)	0.011923	50.29%	0.2190	0.8775

Source: author's own calculations.

The out-of-sample evaluation demonstrates that the regularised linear models did not outperform the naive baseline. For the DXY index, the Diebold-Mariano tests confirm that neither the Ridge nor Lasso regressions significantly improved upon the ARIMA(0,0,0) specification (p -values of 0.2712 and 0.1328, respectively). For the EEM proxy, both specifications similarly failed to improve upon the baseline (p -values of 0.2038 and 0.2190 for Ridge and Lasso, respectively), with the Ridge model additionally recording a directional accuracy below 50% (49.81%), though this deterioration does not reach statistical significance. These results indicate that the macroeconomic feature space provides no exploitable out-of-sample predictive power in either market.

4.2.3 Non-Linear Machine Learning Models

Both the ARIMAX and regularised regression frameworks assume a linear relationship between the predictors and the target returns. However, financial time series are characterised by asymmetric responses to shocks and complex non-linear interactions. To capture these dynamics, the analysis employs three non-linear machine learning algorithms: Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Support Vector Regression (SVR) utilising a radial basis function kernel.

Hyperparameter Tuning and In-Sample Learning

To balance model complexity and prevent overfitting, hyperparameters were optimised using grid search with 5-fold TimeSeriesSplit cross-validation on the training sample. Table 4.10

reports the evaluated parameter grids and the optimal configurations identified for both target assets.

Table 4.10: Hyperparameter Grid Search Space and Optimal Values

Model	Parameter	Search Space	Optimal (DXY)	Optimal (EEM)
RF	n_estimators	100, 200, 500, 1000	500	500
	max_depth	None, 3, 5, 7, 10	5	5
	min_samples_leaf	5, 10, 20, 50	5	5
	min_samples_split	2, 10, 20	2	2
	max_features	sqrt, log2, None	sqrt	sqrt
	bootstrap	True, False	True	True
XGB	n_estimators	25, 50, 100, 200, 300	25	25
	max_depth	2, 3, 5	2	3
	learning_rate	0.01, 0.05, 0.1	0.05	0.05
	min_child_weight	1, 5, 10	5	1
	subsample	0.8, 1.0	0.8	0.8
	colsample_bytree	0.8, 1.0	0.8	0.8
SVR	kernel	rbf	rbf	rbf
	C	1, 10, 100, 1000	1	1
	epsilon	0.001, 0.005, 0.01, 0.05, 0.1	0.05	0.01
	gamma	scale, 0.1, 0.01, 0.001	scale	0.001

Note. Due to computational constraints, the bounds and density of the search grids were kept relatively concise, balancing systematic model tuning with practical feasibility.

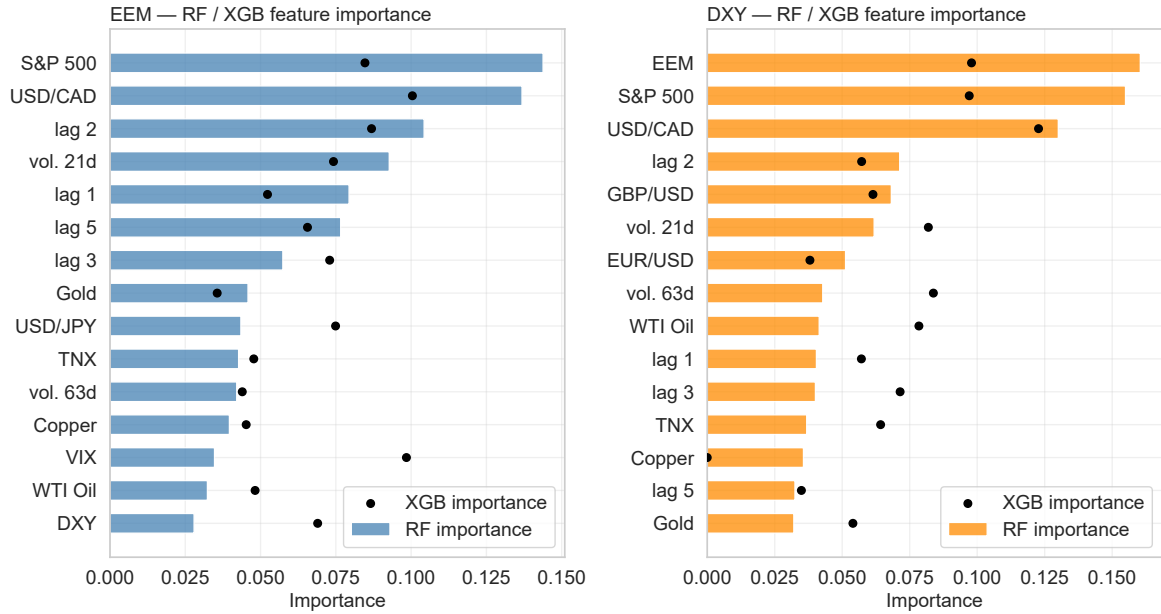
Source: author's own calculations.

An inherent architectural advantage of tree-based models is their ability to compute feature importance scores by measuring the variance reduction attributed to each variable during training. Figure 4.5 plots the normalised impurity-based importance scores extracted from the fitted RF and XGBoost models for both target assets.

The results indicate that cross-asset predictors dominate the feature space for both targets. For the EEM proxy, the S&P 500 index and the USD/CAD exchange rate exhibit the highest importance, followed by the second autoregressive lag and 21-day rolling volatility. This composition suggests a sensitivity of emerging market equities to global risk appetite and short-term momentum. For the DXY index, the EEM proxy and the S&P 500 index are the primary predictors, followed by the USD/CAD exchange rate, which aligns with the linkage of the reserve currency to US equity markets and global capital flows. While the RF and XGBoost algorithms yield broadly consistent importance hierarchies, specific algorithmic divergences exist – most notably, the VIX index emerges as highly influential for the EEM proxy under XGBoost, yet receives marginal weight in the RF model. Nevertheless, this broad alignment suggests that the rankings predominantly reflect the underlying data structure rather than algorithm-specific biases. These findings imply that the non-linear ensembles

utilised the expanded macroeconomic information set during the training phase.

Figure 4.5: Feature Importance: Random Forest vs Gradient Boosting



Source: author's own calculations.

Out-of-Sample Forecasting Performance

Table 4.11 reports the out-of-sample forecasting accuracy of the tuned machine learning algorithms evaluated against the naive baseline.

Table 4.11: Out-of-Sample Performance: Non-Linear Machine Learning Models

Target Asset	Model Specification	RMSE	Dir. Accuracy	DM p -value	Binomial p -value
DXY	Baseline: ARIMA(0,0,0)	0.004706	51.57%	—	0.3241
	Random Forest	0.004693	50.52%	0.2966	0.7580
	Gradient Boosting	0.004694	49.19%	0.1615	0.6220
	Support Vector Reg.	0.004813	46.91%	0.0005*	0.0485 [†]
EEM	Baseline: ARIMA(0,0,0)	0.011834	51.62%	—	0.3089
	Random Forest	0.011972	50.95%	0.0886	0.5580
	Gradient Boosting	0.012000	51.24%	0.0968	0.4409
	Support Vector Reg.	0.012049	49.52%	0.0221*	0.7814

Note. * indicates that the model performed significantly *worse* than the baseline (DM statistic > 0 and p -value < 0.05). [†] indicates that the directional accuracy is significantly *worse* than a random coin toss (Binomial test p -value < 0.05 with accuracy $< 50\%$).

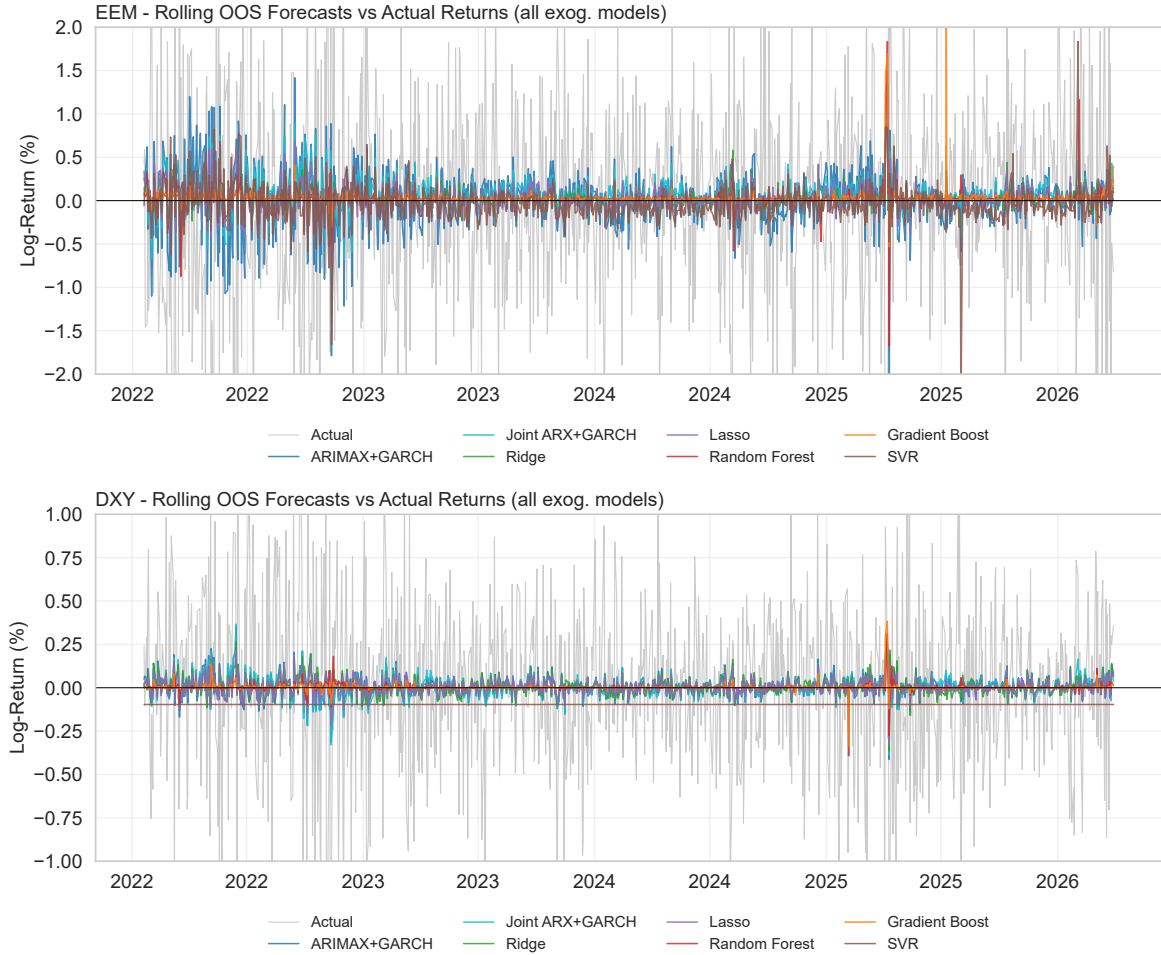
Source: author's own calculations.

The Diebold-Mariano and Binomial tests indicate that none of the machine learning algorithms outperformed the ARIMA(0,0,0) baseline. Moreover, the SVR model significantly

degraded forecasting accuracy for both the DXY index and the EEM proxy.

Figure 4.6 plots the one-step-ahead out-of-sample predictions from all multivariate models against the realised log-returns for both assets. The forecasts predominantly collapse toward the unconditional mean, failing to track the amplitude of the realised return series.

Figure 4.6: Rolling OOS Forecasts vs Actual Returns



Although statistical tests indicate no systematic out-of-sample improvement over the baseline, Figure 4.6 shows that many specifications produce highly smoothed forecast paths that stay close to zero and fail to track the amplitude of realised returns – consistent with weak one-step-ahead predictability. The most extreme case is SVR for DXY, where the predicted series is almost constant; for EEM, Random Forest and Gradient Boosting display especially flat trajectories relative to the actual series, while still varying somewhat over time (unlike the near-degenerate DXY–SVR path). By contrast, models with more volatile forecasts (ARIMAX, joint ARX, Lasso) exhibit greater day-to-day movement, but in our OOS error metrics this does not translate into reliable accuracy gains relative to simpler benchmarks, so the additional variability should be interpreted cautiously rather than as evidence of exploitable signal.

Robustness Validation: Block Permutation Test

To determine whether the machine learning algorithms extract genuine signals rather than overfitting to noise, the analysis employs a non-parametric block permutation test. The exact computational specifications of this procedure are detailed in Section 2.10.

Table 4.12 reports the test results, and Figure 4.7 plots the empirical RMSE distributions.

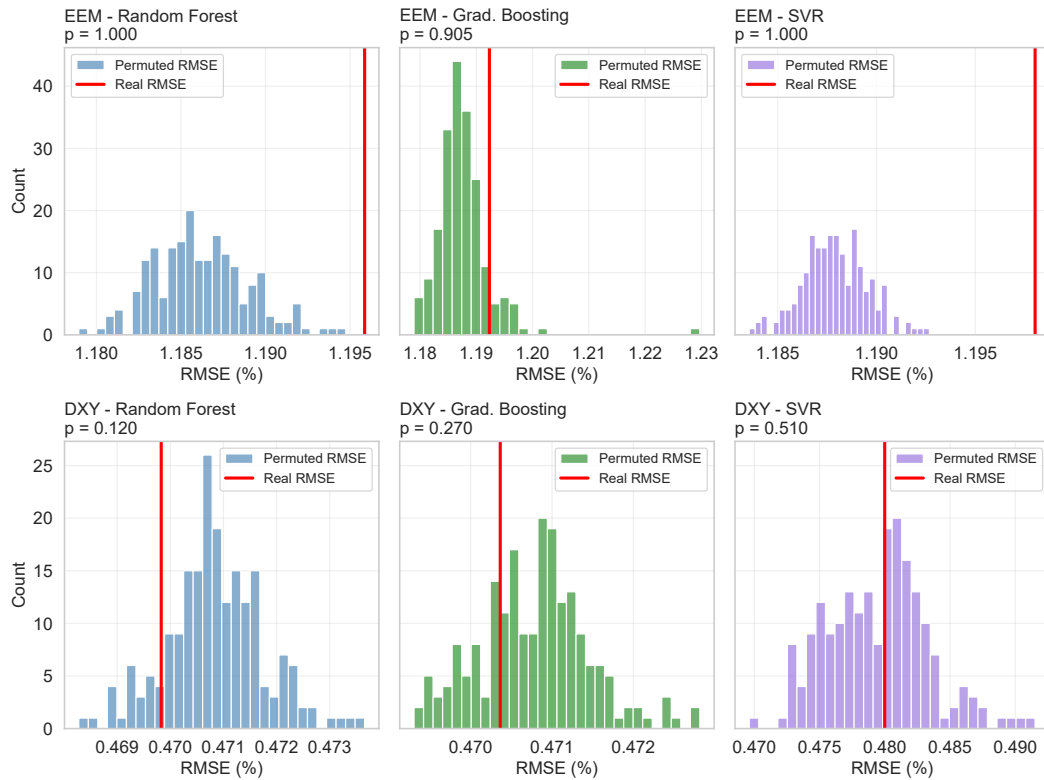
Table 4.12: Block Permutation Test Results (21-day blocks, 200 permutations)

Target Asset	Model	Real RMSE	Mean Permuted RMSE	p -value
DXY	Random Forest	0.004698	0.004709	0.1200
	Gradient Boosting	0.004704	0.004708	0.2700
	Support Vector Reg.	0.004800	0.004795	0.5100
EEM	Random Forest	0.011959	0.011862	1.0000
	Gradient Boosting	0.011923	0.011876	0.9050
	Support Vector Reg.	0.011980	0.011879	1.0000

Note. The empirical p -value represents the probability $P(\text{Permuted RMSE} \leq \text{Real RMSE})$. The slight difference in RMSE compared to Table 4.11 occurs because the training set is block-truncated, and the predictions for the entire test duration are generated from a single model fit.

Source: author's own calculations.

Figure 4.7: Block Permutation Test: Distribution of Permuted RMSEs vs Real RMSE



Source: author's own calculations.

A resulting p -value exceeding 0.05 indicates that any apparent predictive accuracy is primarily attributable to chance. Indeed, the empirical p -values for all machine learning specifications exceed this conventional threshold, meaning the null hypothesis cannot be rejected for either asset. Despite this shared lack of statistical significance, the permutation tests reveal a fundamental divergence in model behaviour. For the DXY index (p -values ranging from 0.1200 to 0.5100), models trained on the authentic temporal sequence marginally outperformed the median block-shuffled permutations. The EEM proxy yields extreme p -values between 0.9050 and 1.0000. These values demonstrate that algorithms trained on pure noise systematically produced lower out-of-sample forecast errors than those trained on actual data, indicating severe overfitting to the authentic sequence.

Consequently, while non-linear algorithms can fit complex in-sample patterns, the permutation tests confirm that these relationships do not generalise out-of-sample. The inability to extract statistically significant predictive signals from the expanded feature space is consistent with the semi-strong form of the Efficient Market Hypothesis. However, as is inherent to standard statistical testing, this failure to reject the null denotes an absence of sufficient evidence for predictability rather than an absolute proof of market efficiency.

4.3 Discussion and Limitations

4.3.1 Comparative Discussion

The in-sample analysis indicates potential differences in the temporal dynamics of the developed currency benchmark (DXY) and the emerging equity market proxy (EEM). As shown in Section 4.1.1, the US Dollar Index exhibits characteristics consistent with a random walk. The Variance Ratio (VR) test values did not significantly deviate from unity, and the Runs test failed to reject the null hypothesis of directional randomness. During the model selection process detailed in Section 4.1.2, the information criteria favoured an AR(0) specification, suggesting a largely memoryless price evolution.

The emerging market proxy exhibits some degree of historical dependence. The formal tests rejected the random walk hypothesis for EEM, with the VR test indicating negative serial correlation at longer lags and the Runs test suggesting potential dependencies in price directions. To account for this, the in-sample mean equation was modelled using an AR(5) specification. Despite differences in the mean equations, exploratory data analysis in Sections 3.3 and 3.4 suggests that both assets experience volatility clustering and leptokurtic tails. This shared characteristic motivated the application of a GARCH(1,1) with a Student's t -distribution.

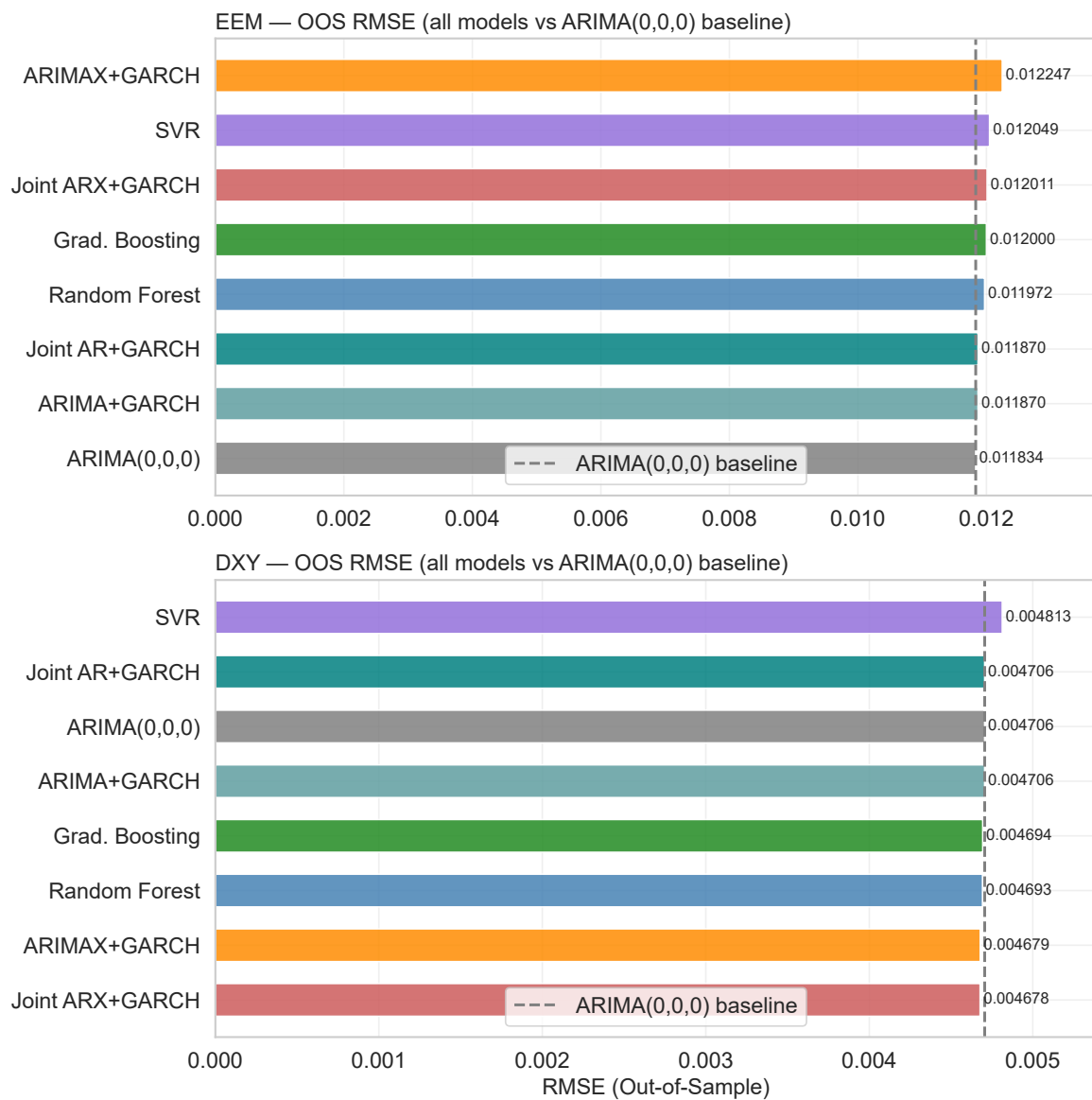
The out-of-sample forecasting results in Sections 4.1.2 and 4.2 suggest that the apparent in-sample patterns in the emerging market might not translate into out-of-sample predictability. Expanding the information set to include exogenous cross-asset predictors yielded comparable

outcomes. The penalised regressions (Ridge and Lasso) and the machine learning algorithms (Random Forest, Gradient Boosting, and SVR) did not outperform the naive baseline. These findings imply that the relationships detected in-sample could be primarily driven by noise.

4.3.2 Evaluation of Hypotheses

With regard to the core research hypotheses defined in Section 2.10, the empirical evidence provides a basis for evaluating market efficiency and predictability. Figure 4.8 summarises the out-of-sample root mean squared errors for all evaluated models relative to the naive baseline.

Figure 4.8: Out-of-Sample RMSE Comparison: All Models vs Naive Baseline



Source: author's own calculations.

Addressing **Hypothesis 1 (Weak-Form Market Efficiency)**, the empirical evidence indicates that neither asset yields exploitable historical patterns. Because the univariate specifications did not reliably outperform the naive baseline out-of-sample, the null hypothesis of weak-form efficiency cannot be rejected for either target market.

Evaluating **Hypothesis 2 (Semi-Strong-Form Market Efficiency)**, the incorporation of exogenous macroeconomic and cross-asset predictors provided no robust out-of-sample advantage. As the multivariate ARIMAX specifications, penalised regressions, and non-linear algorithms did not reliably extract generalisable signals from the expanded information set, the null hypothesis of the semi-strong form of the Efficient Market Hypothesis remains unrejected.

Turning to **Hypothesis 3 (Volatility Modelling)**, the empirical evidence rejects the null hypothesis of constant variance. Both asset classes exhibit conditional heteroskedasticity and volatility clustering. These findings imply that while the directional mean of asset returns lacks out-of-sample predictability, the conditional variance is highly structured and can be modelled using GARCH specifications.

4.3.3 Limitations of the Study

Interpretation of the empirical results requires careful consideration of several methodological constraints inherent to the study’s design. Although the empirical framework is theoretically sound, these boundaries provide necessary context for the conclusions.

While the study deploys non-linear machine learning algorithms (Random Forest, Gradient Boosting, and Support Vector Regression) alongside classical econometrics, this selection does not exhaust the available computational methodologies. Deep learning architectures, particularly Long Short-Term Memory (LSTM) networks and models specifically tailored for financial time series, such as attention mechanisms or deep reinforcement learning, might be better equipped to capture long-term temporal dependencies in sequential data. Beyond architectural choices, computational constraints restricted the hyperparameter search grids to a discrete subset of values. To maintain computational feasibility during the rolling out-of-sample evaluation, this hyperparameter optimisation was executed exclusively once on the initial training partition. Fixing these configurations prevented the algorithms from dynamically adapting their structural complexity to shifting market regimes over time.

Feature space specifications restrict the predictive models to nine exogenous cross-asset variables. For the machine learning specifications, this information set is specifically supplemented by autoregressive lags and rolling volatilities. While the selected commodities, equity indices, and yield proxies capture major macroeconomic cycles, asset prices are simultaneously influenced by an array of omitted fundamental indicators, such as inflation metrics, employment data, and monetary policy aggregates. Market dynamics are increasingly shaped by alternative unstructured data, including sentiment derived from financial news and market microstructure data such as limit order book flows. Excluding this broader information set

restricts the models' capacity to capture sentiment-driven or microstructural anomalies that may exist outside the nine-variable framework.

Daily closing prices serve as the sole temporal frequency for this analysis. It is possible that transient market inefficiencies and predictable patterns exist at higher frequencies, such as intraday or tick-level data. However, such micro-inefficiencies are typically arbitrated away rapidly by high-frequency trading algorithms, potentially rendering them undetectable in daily aggregated timeframes.

Statistical evaluation of the forecasts relies heavily on the Diebold-Mariano test. However, when comparing nested model specifications – such as the baseline ARIMA and the augmented ARIMAX framework – the test statistic is known to exhibit a non-standard asymptotic distribution. Although the application of standard critical values serves as a practical heuristic, this mathematical property introduces a layer of statistical ambiguity when strictly assessing the significance of predictive improvements.

Theoretical ambiguity also arises from the Joint-Hypothesis Problem. Market efficiency is always evaluated jointly with a specific forecasting model. Consequently, the inability of the applied models to detect out-of-sample predictability creates a methodological challenge: it remains uncertain whether the target markets are genuinely efficient, or whether the selected model specifications were inadequately equipped to uncover the true underlying pricing dynamics.

5. Conclusion

This study assessed the weak and semi-strong forms of the Efficient Market Hypothesis (EMH) by comparing two structurally distinct financial environments: the U.S. Dollar Index (DXY) and the iShares MSCI Emerging Markets ETF (EEM). By evaluating these specific assets, the research demonstrated the structural and behavioural differences between a mature global reserve currency and a developing equity market. To conduct this comparative assessment, the methodology combined classical time-series econometrics with more modern, non-linear machine learning algorithms.

The empirical results showed that it was not possible to reject the weak and semi-strong forms of the efficient market hypothesis out-of-sample in this study. Across all evaluated specifications, the advanced models did not statistically demonstrate better out-of-sample predictive power than the naive benchmark. However, this result must be viewed through the lens of the limitations discussed in Section 4.3.3.

Nevertheless, the difference between the two assets is noticeable. The Dollar Index, based on this study, exhibits properties consistent with EMH. This is evident from both the preliminary serial correlation diagnostics and the algorithmic lag selection, which identified a zero-lag specification as optimal. For the EEM, the in-sample analysis rejected the weak form of the EMH. Autocorrelation is significantly more pronounced in the emerging market proxy, leading the information criteria to select higher-order autoregressive structures. However, this in-sample statistical anomaly did not translate into an exploitable predictive signal; the models were unable to outperform the naive benchmark in out-of-sample testing.

Volatility, however, was found to exhibit strong conditional memory for both assets. While the first moment (mean returns) proved unpredictable, the second moment (conditional variance) exhibits clear structural properties. Strong ARCH effects and the adequate fit of the GARCH models demonstrated that, although the future average is unpredictable, the risk magnitude (volatility clustering) can be modelled and used for risk management.

The market hypothesis remains unfalsified for both assets, which supports the theoretical dominance of passive investment strategies in these market segments. This shifts the practical focus from seeking outperformance to the consistent management of uncertainty.

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Appendices

A. Source Code

All source code used in this thesis is publicly available in the following GitHub repository:

<https://github.com/KairosPath/BachelorThesis>

The repository contains the complete implementation of all computational procedures, including data preprocessing, model estimation, forecasting pipelines, and result visualisation.

B. Model Selection and Diagnostics

B.1 GARCH(1,1) Parameter Estimates

Table B.1: Two-Step ARIMA+GARCH Parameter Estimates with Student- t Innovations

Asset	Mean model	$\hat{\omega}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\nu}$
DXY	ARIMA(0,0,0)	0.000746	0.0367	0.9603	10.153
EEM	ARIMA(2,0,1)	0.038261	0.0962	0.8904	8.742

Table B.2: Joint AR-GARCH Parameter Estimates with Student- t Innovations

Asset	Model	$\hat{\omega}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\nu}$
DXY	AR(0)+GARCH(1,1)	0.000747	0.0367	0.9603	10.145
EEM	AR(5)+GARCH(1,1)	0.0396	0.1002	0.8860	8.633

Table B.3: Joint ARX-GARCH Parameter Estimates with Student- t Innovations

Asset	Model	$\hat{\omega}$	$\hat{\alpha}_1$	$\hat{\beta}_1$	$\hat{\nu}$
DXY	ARX(5)+GARCH(1,1)	0.000682	0.0355	0.9617	11.019
EEM	ARX(5)+GARCH(1,1)	0.037209	0.0978	0.8893	8.384

Note. All reported ARCH, GARCH, and Student- t distribution parameters are statistically significant at $p < 0.05$, indicating strong volatility clustering, high persistence, and heavy-tailed innovations in both return series.

B.2 ARX-GARCH Model Selection

Table B.4: Joint ARX-GARCH AIC Ranking within the Main Search Range

Asset	Model	AIC	Δ AIC
DXY	ARX(5)+GARCH(1,1)	4918.29	0.00
DXY	ARX(3)+GARCH(1,1)	4918.71	0.42
DXY	ARX(1)+GARCH(1,1)	4919.25	0.96
EEM	ARX(5)+GARCH(1,1)	14716.03	0.00
EEM	ARX(4)+GARCH(1,1)	14720.50	4.47
EEM	ARX(3)+GARCH(1,1)	14723.58	7.55

C. Coefficient Estimates

C.1 Two-Step ARIMAX Coefficient Estimates

Table C.1: Two-Step ARIMAX+GARCH Coefficient Estimates

Target Asset	Variable	Coefficient	Std. Error	p-value
DXY	MA(1)	−0.0304	0.0154	0.0480
	Constant	0.0000	0.0001	0.5470
	MSCI EM (EEM)	−0.0140	0.0057	0.0140
	Copper (HG=F)	0.0069	0.0039	0.0800
	10Y Treasury (^TNX)	0.0102	0.0021	<0.001
	Gold (GC=F)	0.0003	0.0057	0.9560
	WTI crude (CL=F)	−0.0009	0.0026	0.7140
	EUR/USD (EURUSD=X)	−0.0047	0.0094	0.6170
	GBP/USD (GBPUSD=X)	−0.0284	0.0124	0.0220
	USD/CAD (CAD=X)	0.0022	0.0129	0.8620
	S&P 500 (^GSPC)	−0.0367	0.0090	<0.001
EEM	AR(1)	−0.5250	0.0590	<0.001
	AR(2)	−0.0238	0.0129	0.0650
	MA(1)	0.5972	0.0580	<0.001
	Constant	0.0006	0.0005	0.1880
	US Dollar Index (DX-Y.NYB)	−0.0065	0.0557	0.9070
	CBOE Volatility (^VIX)	−0.0137	0.0048	0.0040
	10Y Treasury (^TNX)	−0.0154	0.0075	0.0390
	WTI crude (CL=F)	−0.0178	0.0096	0.0640
	Copper (HG=F)	0.0215	0.0112	0.0560
	Gold (GC=F)	−0.0281	0.0179	0.1170
	S&P 500 (^GSPC)	−0.4003	0.0265	<0.001
	USD/CAD (CAD=X)	−0.1014	0.0396	0.0110
	USD/JPY (JPY=X)	−0.1079	0.0260	<0.001

Note. Coefficients from the two-step estimation procedure.

C.2 Joint ARX+GARCH Coefficient Estimates

Table C.2: Joint ARX(5)+GARCH(1,1) Mean-Equation Coefficient Estimates

Target Asset	Variable	Coefficient	Std. Error	<i>t</i>	<i>p</i> -value
DXY	Constant	0.0000	0.0001	0.490	0.6240
	AR(1)	−0.0416	0.0186	−2.240	0.0250
	AR(2)	−0.0090	0.0192	−0.471	0.6380
	AR(3)	−0.0195	0.0151	−1.290	0.1970
	AR(4)	−0.0126	0.0155	−0.810	0.4180
	AR(5)	0.0077	0.0154	0.504	0.6140
	MSCI EM (EEM)	−0.0102	0.0080	−1.279	0.2020
	Copper (HG=F)	0.0038	0.0052	0.736	0.4620
	10Y Treasury ($\hat{\text{TNX}}$)	0.0106	0.0038	2.819	0.0048
	Gold (GC=F)	−0.0085	0.0082	−1.042	0.2970
	WTI crude (CL=F)	−0.0025	0.0031	−0.818	0.4130
	EUR/USD (EURUSD=X)	−0.0097	0.0158	−0.614	0.5390
	GBP/USD (GBPUSD=X)	−0.0253	0.0146	−1.736	0.0830
	USD/CAD (CAD=X)	−0.0111	0.0161	−0.693	0.4880
	S&P 500 ($\hat{\text{GSPC}}$)	−0.0321	0.0124	−2.582	0.0098
EEM	Constant	0.0010	0.0002	5.203	<0.001
	AR(1)	0.0305	0.0273	1.117	0.2640
	AR(2)	−0.0137	0.0176	−0.779	0.4360
	AR(3)	−0.0155	0.0161	−0.965	0.3350
	AR(4)	−0.0304	0.0159	−1.919	0.0549
	AR(5)	−0.0334	0.0157	−2.127	0.0335
	US Dollar Index (DX-Y.NYB)	−0.0369	0.0524	−0.703	0.4820
	CBOE Volatility ($\hat{\text{VIX}}$)	−0.0105	0.0045	−2.343	0.0191
	10Y Treasury ($\hat{\text{TNX}}$)	−0.0126	0.0100	−1.266	0.2050
	WTI crude (CL=F)	−0.0020	0.0095	−0.215	0.8300
	Copper (HG=F)	0.0246	0.0154	1.597	0.1100
	Gold (GC=F)	−0.0399	0.0234	−1.703	0.0886
	S&P 500 ($\hat{\text{GSPC}}$)	−0.2541	0.0464	−5.477	<0.001
	USD/CAD (CAD=X)	−0.0726	0.0474	−1.531	0.1260
	USD/JPY (JPY=X)	−0.0390	0.0317	−1.231	0.2180

Note. Coefficients from the joint ARX(5)+GARCH(1,1) specification based on joint maximum likelihood estimation. Standard errors are robust to conditional heteroskedasticity.

D. Data Characteristics

D.1 Exogenous Variable Stationarity Tests

Table D.1: ADF and KPSS Stationarity Tests for Exogenous Predictors

Variable	ADF t -stat	ADF p -value	KPSS τ_μ	KPSS p -value
Copper (HG=F)	−19.872	< 0.001	0.090	> 0.10
10Y Treasury (^TNX)	−12.020	< 0.001	0.155	> 0.10
Gold (GC=F)	−24.689	< 0.001	0.209	> 0.10
WTI crude (CL=F)	−12.736	< 0.001	0.063	> 0.10
EUR/USD (EURUSD=X)	−13.479	< 0.001	0.038	> 0.10
GBP/USD (GBPUSD=X)	−15.789	< 0.001	0.049	> 0.10
USD/CAD (CAD=X)	−14.697	< 0.001	0.159	> 0.10
S&P 500 (^GSPC)	−18.378	< 0.001	0.138	> 0.10
CBOE Volatility (^VIX)	−28.832	< 0.001	0.017	> 0.10
USD/JPY (JPY=X)	−13.054	< 0.001	0.207	> 0.10

D.2 Descriptive Statistics for Exogenous Predictors

Table D.2: Descriptive Statistics for Exogenous Predictors (% log-returns)

Variable	Mean	Std. Dev.	Skewness	Kurtosis	Min	Max
Copper (HG=F)	0.026	1.780	−0.607	14.11	−25.17	12.44
10Y Treasury (^TNX)	0.004	2.573	0.205	34.72	−34.70	40.48
Gold (GC=F)	0.044	1.157	−0.544	10.11	−12.07	8.64
WTI crude (CL=F)	0.024	2.623	0.075	21.14	−28.22	31.96
EUR/USD (EURUSD=X)	0.000	0.703	0.587	106.85	−14.33	15.96
GBP/USD (GBPUSD=X)	−0.007	0.587	−0.689	12.56	−7.91	3.48
USD/CAD (CAD=X)	0.003	0.536	0.202	6.56	−3.69	3.60
S&P 500 (^GSPC)	0.034	1.210	−0.477	16.22	−12.77	10.96
CBOE Volatility (^VIX)	0.010	7.595	1.038	9.52	−44.24	76.82
USD/JPY (JPY=X)	0.008	0.737	−0.588	112.70	−16.85	16.29

Note. Sample period: 2005-05-02 to 2026-03-31. All descriptive statistics are computed on the log-return series.

E. AI Tools Usage

In accordance with the declaration provided at the beginning of this thesis, this appendix details the specific applications of AI-assisted tools during the preparation of the study “Testing Market Efficiency: Evidence from the US Dollar Index and Emerging Markets”.

Tools Used

Perplexity AI – used for academic literature search and discovery of relevant sources. All identified sources were independently located, read, and verified by the author before citation.

Claude Sonnet (Anthropic) – used in a supplementary capacity for the following purposes:

- Proofreading and language refinement of written sections
- Improving stylistic quality and clarity of academic expression
- Generating parts of the source code (primarily visualisation scripts and chart formatting)
- Debugging and correcting errors in Python code
- Reviewing the logical structure and flow of chapters
- Adding explanatory markdown annotations to Jupyter Notebook cells
- Generating interactive reports based on statistical test outputs

Scope of AI Use

All analytical work, econometric modelling, interpretation of results, and conclusions are entirely the author’s own. AI tools did not perform any statistical analysis or make any research decisions. All AI-generated content was reviewed and verified by the author before inclusion in the thesis.